

Targeting Humanitarian Food Assistance: A Field Experiment in Eastern DRC*

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Abstract

Targeting in social protection is well studied, but evidence on humanitarian targeting of transient needs amid widespread vulnerability remains limited, particularly in fragile settings. We implement a cluster-randomized trial in Tanganyika Province, Democratic Republic of the Congo, comparing data-driven targeting (PMT Plus) with community-based targeting (CBT) by local committees for humanitarian food assistance. We assess targeting precision, coverage, welfare, perceived legitimacy, and community relations. CBT reduces coverage by 11 percentage points without compromising precision or welfare outcomes among beneficiaries or non-beneficiaries. Satisfaction with targeting, social cohesion, and perceived insecurity are statistically indistinguishable across arms. Adopting a more conflict-sensitive CBT could therefore help expand program coverage under funding constraints.

Keywords: Targeting, Humanitarian assistance, Fragile setting, Community-based targeting, Data-driven testing, Welfare impacts, Social cohesion.

JEL Codes: C93, I38, O12, O17, D63.

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1 Introduction

Governments and international organizations use various methods to identify beneficiaries of social protection and humanitarian assistance. In both settings, policymakers observe imperfect signals of household vulnerability and face binding budget constraints that prevent them from assisting everyone in need (Banerjee et al., 2024). In humanitarian settings, however, these challenges are particularly acute because need is widespread and shocks create temporarily vulnerable households that conventional targeting methods may struggle to identify (Del Ninno and Mills, eds, 2015; Costella and Ovadiya, 2014; Schnitzer, 2019). The same shocks may also weaken social ties and heighten tensions, increasing the social costs of selective inclusion and exclusion (Della Guardia et al., 2022; Stys et al., 2026). Despite these differences, the existing targeting evidence draws primarily on social-protection programs¹ and focuses on targeting accuracy (Cameron and Shah, 2014), offering limited insights into whether its conclusions extend to humanitarian settings or, more fundamentally, whether targeting itself is appropriate under these conditions (Jeong and Trako, 2022; Devereux et al., 2017).

Among targeting methods, Proxy Means Testing (PMT) and Community-Based Targeting (CBT) are two of the most widely used approaches in social protection that have also been adopted in large-scale humanitarian operations. PMTs are often promoted as a more objective tool (Follett and Henderson, 2023), and existing evidence suggests that they may be more efficient in identifying consumption-poor households (Premand and Schnitzer, 2021; Brown et al., 2018). However, PMTs have important limitations. They often require costly census-style data collection exercises, and their performance, in addition to being affected by measurement error inherent to statistical models (Brown et al., 2018; Banerjee et al., 2024), is highly sensitive to the quality and timeliness of the data on which they rely (Follett and Henderson, 2023; Sharif, 2012).

¹See Coady et al. (2004), Grosh et al., eds (2022), and Devereux et al. (2017) for targeting reviews of social protection and anti-poverty programs.

PMTs are also often perceived as difficult to explain to recipients, which may give rise to concerns about limited transparency and the fairness of allocation (Cameron and Shah, 2014; Hanna and Olken, 2018). By contrast, community-based targeting (CBT) is generally easier to communicate and may foster a greater sense of participation and local ownership (Alatas et al., 2012). Nonetheless, CBT has been associated with risks of elite capture and favoritism (Alatas et al., 2019; Han and Gao, 2019). The trade-offs between these approaches, balancing concerns about fairness and legitimacy under PMT against the risks of misallocation and broader social consequences under community-based targeting, remain poorly understood, despite being central to the design of humanitarian programs (Devereux et al., 2017).

To provide some of the first experimental evidence from humanitarian settings, and among the most comprehensive assessments to date of both the targeting precision and broader social trade-offs of PMT and community-based targeting, we partnered with the World Food Programme (WFP) to implement a cluster-randomized trial in the Democratic Republic of the Congo (DRC), home to one of the world’s largest humanitarian crises. We randomized 96 blocs (communities) participating in a large humanitarian cash assistance program (about USD 54 per person, roughly 70% of median per capita expenditure over the six-month cycle) to either PMT+ (the status quo approach) or CBT in Tanganyika Province. Prior to program implementation, we conducted a census and subsequently fielded two rounds of follow-up surveys: the first immediately after the final transfer, six months after program start, and the second eight months later. Reflecting the operational challenges of working in this context, follow-up data were collected in 84 of the original 96 blocs due to security-related operational constraints. Nevertheless, the randomization design, stratified by quadruplets of similar cluster size, and the preserved balance across the 84 observed blocs support a range of robustness checks and bolster confidence in our findings.

Our first finding concerns coverage, defined as the share of households selected for assistance, a margin that most targeting studies hold constant across methods. We benchmark coverage against the prevalence of food insecurity, defined in our setting, as in many human-

itarian programs, by a Food Consumption Score (FCS) of 28 or lower. By this measure, 86% of households are food insecure. Both methods select fewer households than the share classified as food insecure under our benchmark. PMT+ selects 80% of households, compared with 69% under CBT. This 11 percentage point (p.p.) coverage gap is operationally large and statistically significant at the 1% level.

In terms of targeting precision, defined as the ability to include all and only food insecure households under the same benchmark, PMT+ is slightly more precise than CBT because it leaves out fewer food insecure households, and thus has a lower exclusion error. However, this difference is explained by its higher coverage rate. When coverage is held constant, exclusion errors under the two methods become virtually identical. Alternative precision measures that are more sensitive to the distribution, including the Coady–Grosh–Hoddinott index and the Targeting Differential, also show no meaningful differences between the two approaches, even when an wealth based wealth index is used as the benchmark.

The characteristics of selected households reveal that the two methods prioritize different dimensions of vulnerability. CBT places greater weight on agricultural and seasonal vulnerability, whereas PMT+ places greater weight on demographic characteristics such as illiteracy, displacement, orphanhood, and old age. When selection rates are compared with food insecurity within each subgroup, both methods undertarget most groups because coverage falls below measured need. CBT provides greater coverage to households with a pregnant or lactating woman and to those headed by a person with a disability, although the latter difference is less precisely estimated. PMT+ instead overrepresents households with an illiterate head, with this difference significant at the 1% level.

Another distinction from the existing targeting literature is that we experimentally estimate downstream outcomes across the entire target population, including beneficiaries and nonbeneficiaries. Despite their different coverage rates, PMT+ and CBT produce similar average FCS, coping strategies, and life satisfaction. These results are unchanged after adjusting for coverage, baseline inequality, and measured need. Outcomes among nonbenefi-

ciaries are also similar across methods, including in blocs where CBT generated a larger gap between coverage and food insecurity. The absence of differences between targeting methods does not mean that the assistance itself was ineffective. Receipt of assistance increased FCS by 3.0 points at the first follow-up and 1.9 points at the second relative to nonbeneficiaries, with both effects significant at the 1% level. Thus, assistance improved recipients' food consumption, but differences in coverage and targeting precision did not translate into detectable differences in average welfare across the target population.

Perceptions of the targeting process and social outcomes are broadly similar across arms. We find no significant differences in overall satisfaction with targeting, social cohesion, or community tensions. Some individual items suggest that households in CBT blocs view the process as slower and marginally less fair. We also find no evidence of favoritism, as knowing or being related to a committee member does not increase the probability of selection.

Despite widespread need, targeting is likely to remain the *de facto* approach in humanitarian settings, both because resources are scarce and because donors and communities hold strong views about who deserves assistance (Banerjee et al., 2024). The relevant choice for social planners may therefore be not whether to target, but which method can produce similar welfare outcomes at the lowest economic and social cost, freeing resources to reach communities that would otherwise remain unserved.

Having established that PMT+ and CBT produce similar average welfare outcomes across different levels of baseline need and inequality, we turn to costs. Implementation costs were nearly identical across arms, at USD 61,881 for PMT+ and USD 62,570 for CBT, since both methods used the same census for the experiment. Targeting cost USD 10.6 per household screened, only 1.4% of the USD 774 household transfer. Screening roughly 73 households therefore costs the same as funding one additional transfer. The larger potential savings arise from differences in coverage. Under a fixed budget of USD 10 million, CBT's lower caseload per bloc would allow the program to reach 282 blocs, compared with roughly 200 under universal coverage. By delivering similar average welfare outcomes at lower total cost,

with limited adverse effects, CBT may offer a viable alternative to navigate funding cuts and constraints in the current fiscal environment.

Our study contributes to several strands of the targeting literature. First, we contribute to research comparing community based and data driven targeting. Previous studies in Indonesia, Cameroon, and rural Burkina Faso find that PMT identifies consumption poor households more accurately than community based methods (Alatas et al., 2012; Stoeffler et al., 2016; Hillebrecht et al., 2020).² This advantage is less consistent when performance is assessed against food security. Stoeffler et al. (2016) find lower inclusion but higher exclusion errors under community targeting, while Premand and Schnitzer (2021) and Schnitzer and Stoeffler (2024) find that PMT’s overall advantage weakens. We extend this literature by allowing each method to determine its own coverage, as occurs in practice, and by randomly assigning the targeting method to estimate downstream outcomes. These features distinguish our study from Premand and Schnitzer (2021) and Schnitzer and Stoeffler (2024). We also extend the experimental comparison in Alatas et al. (2012) to an active conflict humanitarian setting.

Evidence on how targeting impact downstream outcomes remains limited and mixed, even though targeting methods select households with different propensities to consume, invest, and share assistance (Haushofer et al., 2025). Alatas et al. (2012) find similar outcomes under PMT and community based targeting, whereas Basurto et al. (2020) find that communities select households that use fertilizer subsidies more effectively. Premand and Schnitzer (2021) find similar expenditure and FCS across methods but higher food stocks under PMT. We add to this evidence by showing that CBT can produce similar average welfare outcomes at lower total expenditure, making it a potentially useful tool for allocating scarce resources amid large aid cuts (Banerjee et al., 2024).

²A related literature measures the quality of the information communities hold rather than comparing methods at the program level: Beaman et al. (2021) ask residents of Monrovia to identify poor neighbors, and Trachtman et al. (2026) measure how much households know about their neighbors’ welfare in Indonesia. Basurto et al. (2020) compare chief led targeting of fertilizer subsidies with a PMT in Malawi, a leader based rather than committee based process.

Experimentally studying how targeting methods jointly affect coverage and welfare allows us to ask whether the choice of method changes not only who receives assistance, but also how many households are reached and whether these differences translate into welfare effects. This question is especially salient under tight budgets and in light of evidence on the spillover and multiplier effects of assistance (Angelucci and De Giorgi, 2009; Delius and Sterck, 2024; Egger et al., 2022; Filmer et al., 2023). Such effects imply that welfare depends on both the households selected and the share of each community receiving assistance. Understanding these two margins is therefore important for allocating scarce humanitarian resources. The closest antecedent is Abay et al. (2026), a cluster randomized experiment in conflict affected Ethiopia that also imposes no quota. That study examines how communities trade off coverage and transfer size as budgets vary, whereas we examine how the choice of targeting method determines coverage and downstream household welfare. To our knowledge, this is the first experiment to study both outcomes jointly.

Finally, we contribute by moving beyond the traditional focus on targeting precision to examine broader tradeoffs that may be especially relevant when assistance requires selective inclusion amid widespread need and fragility. Evidence on the social consequences of CBT remains limited and mixed. Incorporating local views of deservingness may increase acceptance, but it may also create favoritism, discrimination, and conflict (Schnitzer and Stoeffler, 2024). Alatas et al. (2012), Sumarto et al. (2025), and Aiken et al. (2025) find that communities prefer community based targeting to PMT. In contrast, Premand and Schnitzer (2021) find similar preferences among beneficiaries and weaker support for community based targeting among nonbeneficiaries, even though the latter perceive it as producing fewer errors. In our setting, satisfaction with the targeting process, social cohesion, and community tensions remain similar across methods despite substantial differences in coverage.

The paper is organized as follows. Section 2 describes the setting, the assistance program, and the experimental design. Section 3 describes the data, the food-security benchmark against which we evaluate targeting, and the study timeline. Section 4 sets out the empirical

strategy—showing why endogenous coverage and the choice of welfare benchmark drive the comparison—and establishes the study’s internal validity. Section 5 presents the results—on coverage, precision, the profiles of selected households, welfare, and community outcomes. Section 6 turns to the cost of targeting and its policy implications, and Section 7 concludes.

2 Context and Experimental Design

2.1 Setting

The Democratic Republic of the Congo (DRC) faces one of the world’s largest and most protracted humanitarian crises. Driven primarily by armed conflict and compounded by climate-related and other shocks, the crisis has displaced approximately five million people, strained host communities, and disrupted markets, agricultural production, and livelihoods, deepening poverty, food insecurity, and malnutrition (IPC, 2025; FSIN and GNAFC, 2025; IDMC, 2024). An estimated 27.7 million people nationwide face acute food insecurity and are considered in need of humanitarian food assistance (IPC, 2025).³ The humanitarian crisis is particularly acute in four eastern provinces—Tanganyika, the setting of our study, North Kivu, South Kivu, and Ituri—where armed groups have long contested land, resources, and political authority (Coghlan et al., 2006; Autesserre, 2010). In 2025, these provinces accounted for 10.3 million of the 27.7 million people facing acute food insecurity nationwide (IPC, 2025).⁴

Despite the scale and severity of the crisis, the humanitarian response in the DRC remains among the world’s most underfunded (OCHA, 2026), forcing humanitarian actors to make difficult decisions about how to allocate scarce resources. An estimated 26–27 million people

³The Integrated Food Security Phase Classification (IPC) ranks acute food insecurity from Phase 1 (Minimal) to Phase 5 (Famine). Phase 3 (Crisis) or above indicates food-consumption gaps, elevated acute malnutrition, or the depletion of essential livelihood assets and marks the threshold at which humanitarian assistance is typically warranted. See https://www.ipcinfo.org/ipc/technical/manual_en.

⁴The DRC is divided into provinces, which are subdivided into territories; territories into sectors and chiefdoms; and these in turn into groupements, the lowest administrative tier and the level within which the communities in our study are located.

required food assistance between 2021 and 2024, yet the number reached by WFP, the country’s largest humanitarian actor, fell from 6.3 million in 2021 to 5.2 million in 2024 and 3.8 million in 2025.⁵ These allocation decisions take place amid protracted conflict, including widespread sexual and gender-based violence, that has eroded trust, polarized intergroup relations, and weakened social cohesion (Johnson et al., 2010; Bartels et al., 2010; Longombe et al., 2008; Vinck et al., 2008; Pham et al., 2022). The challenge is therefore not only to identify people in need, but also to determine whom and how best to prioritize in an already fragile social environment.

Stoeffler et al. (2020) review beneficiary selection across 69 programs in five regions of the DRC and find that community-based selection is already a dominant practice: 57 programs involve the community in some role. Its performance, however, is almost never measured. Organizations judge their own targeting by complaints and anecdote rather than by error rates—50 of the 69 reported zero inclusion and exclusion errors—and rarely look beyond selection itself, leaving effects on welfare and community relations unexamined.

2.2 WFP’s General Food Assistance (GFA) Program

Our study is embedded in WFP’s General Food Assistance (GFA) program, which provides unconditional in-kind or cash assistance to eligible households in communities facing severe food insecurity. Humanitarian food assistance is a temporary emergency response rather than a standing social protection entitlement. It addresses acute consumption shortfalls caused by shocks such as displacement, flooding, and seasonal food insecurity, with support expanding or contracting as needs change. Social protection instead provides more predictable, system based support against chronic poverty and vulnerability (Sabates-Wheeler et al., 2022). GFA follows this emergency response model. Assistance is delivered in cycles of three to six months during periods of greatest need, with caseloads reassessed at the start of each cycle as needs

⁵Estimates of need come from IPC analyses covering different periods between February 2021 and June 2024 (IPC, 2021a,b, 2023, 2024). Beneficiary totals come from WFP’s Annual Country Reports (WFP, 2024, 2025a).

and available funding change (WFP, 2025b). Persistent funding shortfalls require WFP to prioritize among far more households than it can assist. Transfer values are based on the minimum expenditure basket (MEB).⁶

We leverage the expansion of the GFA program to a set of blocs—clusters of geographically proximate households with established social ties—located within three groupements in Tanganyika Province. The expansion took place during the May–October 2023 program cycle and provided assistance through cash transfers delivered in four tranches. For the first three payments, delivered between May and August 2023, households received 30,000 Congolese francs per household member (approximately USD 13 at the nominal exchange rate or USD 31 in 2023 PPP). In September 2023, WFP increased the final payment to 36,000 francs per household member (approximately USD 15 nominal or USD 38 in PPP) following an update to the MEB.⁷

The amount of cash assistance received was intended to cover a six-month period. Total transfers amounted to about 126,000 francs per household member (equivalent to USD 54 or USD 132 in 2023 PPP dollars), or about 70% of median per capita expenditure in Tanganyika over the same six-month period.⁸ Transfers were delivered in cash at designated distribution points to households registered as beneficiaries in WFP’s beneficiary registration platform, SCOPE.

WFP’s protocol to target their GFA program involves two steps. First, it uses data from the Integrated Food Security Phase Classification (IPC) to identify geographic areas at risk of severe hunger crises. Following geographic prioritization, WFP conducts a more granular targeting at the household level to identify beneficiaries. Our evaluation focuses on this second step and compares two targeting approaches: proxy means test (PMT)-based

⁶The MEB represents the per capita cost of meeting basic food and non-food needs, as agreed by the country’s Cash Working Group (Cash Working Group, 2021).

⁷Nominal USD values use an exchange rate of 2,340 Congolese francs (CDF) per USD. PPP values use a 2023 conversion factor of 957.73 CDF per international dollar (World Bank, World Development Indicators).

⁸Median per capita expenditure in Tanganyika is 30,130 CDF per month, against a monthly transfer of 21,000 CDF per member; measured against the mean the transfer represents 60%. Computed from the WFP Emergency Food Security Assessment (EFSA) 2023, restricted to Tanganyika. Appendix Table A1 reports the underlying figures.

targeting, WFP’s current targeting approach implemented for all GFA in the country, and a newly introduced community-based targeting (CBT) alternative.

2.3 Experimental Design

We use a stratified cluster randomized controlled trial, with the bloc serving as both the experimental cluster and the unit of randomization. Blocs are similar to the units studied by [Alatas et al. \(2012\)](#). Their scale facilitates the formation of local targeting committees and makes them the natural unit for beneficiary selection and program delivery.

Before randomization, WFP’s implementing partners conducted scoping visits across the study area to enumerate all blocs and estimate their populations. They identified 165 blocs comprising approximately 13,600 households. Of these, 96 met the prespecified eligibility criteria for inclusion in the impact evaluation: (i) *no WFP assistance during the preceding 12 months*, (ii) *classification as experiencing crisis-level food insecurity*, and (iii) *a population of at least 40 households*. We grouped the 96 eligible blocs into quadruplets based on geography and population size, with each quadruplet forming a randomization stratum. Within each stratum, we randomly assigned blocs to either PMT+, WFP’s current approach to household targeting, or the community based targeting (CBT) alternative. Appendix Figure [A1](#) summarizes the experimental design and participant flow. The two methods select beneficiaries as follows:

Arm 1 — PMT+ (status quo approach). The PMT “Plus” approach was the method WFP used to target General Food Assistance (GFA) at the time of the study. It has two core elements: (i) a data-driven model to predict household food insecurity (PMT), and (ii) “lock-in” criteria selected through focus groups (the “plus” element). The PMT component identifies food-insecure households, defined as those with a Food Consumption Score (FCS) at or below 28.⁹ We trained the PMT model on prior WFP Post-Distribution Monitoring

⁹The FCS is a composite indicator developed by the World Food Programme (WFP) that captures dietary diversity and food frequency using weighted food groups consumed over the previous seven days.

(PDM) surveys from the same context, which record both the outcome of interest (FCS) and a set of candidate predictors. Following evidence that machine learning approaches such as LASSO can improve targeting performance (e.g., [Cao et al., 2025](#); [Altındağ et al., 2021](#)), we tested several alternatives and found that a LASSO-based procedure delivered the best out-of-sample predictive performance. We therefore use it to select predictors and estimate model parameters. We then applied the estimated model to census data the CO collected between December 2022 and March 2023, generating predicted FCS for all households and deeming those at or below 28 eligible.

The “**Plus**” component consists of additional inclusion criteria elicited through gender-separated focus group discussions (FGDs) conducted in three randomly selected blocs. We recorded all household characteristics that participants identified as indicators of vulnerability and mapped them onto measures available in the household questionnaire. Several were observed directly, including the presence of elderly household members, female household headship, and household members with disabilities. For characteristics that were not measured directly, including adolescent motherhood and household heads who were pregnant or breastfeeding, we constructed proxies using the available data.

The FGDs identified eight criteria that were common across blocs. We use these criteria as “lock-in” rules: a household is included regardless of its PMT score if it satisfies at least one of them.¹⁰ All eight criteria were included in the questionnaire ex ante, based on insights from earlier consultations.

The eight criteria are: (i) households headed by a woman with no adult male engaged in an income-generating activity; (ii) households headed by an elderly person, defined as someone aged 60 or older; (iii) households headed by a person with a disability or chronic

Standard thresholds classify households as having poor (FCS < 28), borderline (28.5–42), or acceptable (> 42) food consumption, with adjustments for high-sugar contexts ([WFP, 2008](#)).

¹⁰Although tailored to the study population, these criteria are broadly consistent with standard definitions of “persons at risk” in humanitarian settings. See, for example, the United Nations High Commissioner for Refugees (UNHCR) Emergency Handbook ([UNHCR, 2018](#)) and the IASC Operational Guidelines on the Protection of Persons in Situations of Natural Disasters ([Brookings-Bern Project on Internal Displacement, 2011](#)).

illness; (iv) households with orphaned children; (v) households with an adolescent mother; (vi) households with a pregnant or lactating woman; (vii) households headed by a widower; and (viii) displaced households. Applying the “**Plus**” criteria increased the number of selected beneficiary households by 29%.

Arm 2 — CBT (community-based alternative). The community-based targeting (CBT) approach delegates the definition of vulnerable household to local committees. In each CBT bloc, WFP’s cooperating partners convened residents to select a 12-member targeting committee comprising six women and six men. Committee membership was chosen to reflect variation in age, ethnicity, and disability status and to ensure the representation of both younger and older residents. Village chiefs and members of their immediate families were ineligible to serve, and committee membership remained fixed after its formation. In total, 42 committees were established, one in each CBT bloc.

Facilitators asked each committee to identify the household characteristics that indicate the greatest need for food assistance. Committees were required to specify at least five criteria, and any household satisfying at least one criterion was deemed eligible. On average, committees selected nine criteria, with the number ranging from five to thirteen. Across the 42 blocs, they identified 42 distinct criteria. The two most common criteria—each selected by 91% of committees—were female-headed households with no adult male engaged in an income-generating activity and households headed by an elderly person. Two other criteria adopted by a majority of committees were not included among the PMT+ lock-in rules: belonging to the Twa ethnic group (56%) and having any household member with a disability (51%). Some committees also selected criteria capturing recent shocks, such as crop loss (12%). In practice, we constructed the beneficiary list for each bloc by applying its committee’s criteria to the same census data used for PMT+.

The distinction between the two arms lies in how eligibility is determined from the same census data. Under PMT+, all communities are assessed using a common statistical model

and the same set of lock-in criteria, derived from three focus group discussions. Under CBT, each community establishes its own eligibility criteria. Importantly, neither arm was given a beneficiary quota, so each method’s own definition of need determines its caseload—the feature we exploit throughout.

2.4 Sampling

Twelve of the 96 study blocs, six in each arm, were dropped before the study began because of security concerns, leaving a final sample of 84 blocs that remains balanced¹¹. Within each remaining bloc we surveyed both beneficiary and non-beneficiary households, taking all households in whichever group was smaller and drawing the remainder at random from the other. The resulting evaluation sample comprises 3,998 households—2,037 in PMT+ blocs and 1,961 in CBT blocs—drawn from a study population of 11,732 households across 84 blocs. Non-beneficiaries constituted the smaller group in most blocs and were therefore oversampled relative to their share of the study population. Of the surveyed households, 2,694 were eligible for cash transfers and 1,304 were ineligible (see evaluation sample in Appendix Table A2).

3 Data and Timeline

3.1 Data

We combine three data sources. First, a baseline census covers the entire target population and provides the food-security benchmark used to evaluate targeting performance. Second, WFP administrative records document transfer receipt, grievances, and program costs. Third, two rounds of household surveys measure outcomes in the evaluation sample. We describe each source below.

¹¹The 12 dropped blocs comprised six from each study arm. Because they exited before randomization and baseline data collection, their removal is orthogonal to treatment assignment and does not bias the census-based precision measures; we confirm overall balance in Section 4.2.

Baseline census. Before beneficiary selection and transfer disbursement, WFP conducted a census of all 11,732 households in the 84 study blocs. The census collected demographic characteristics, asset ownership, housing conditions, access to services, and seven-day food consumption. These data provide household characteristics used in targeting, allow us to evaluate targeting performance across the full study population rather than a sample, and supply baseline covariates for the econometric analysis.

Food-security benchmark. Our objective is to evaluate how well each method targets food insecurity, which we benchmark using the Food Consumption Score (FCS). FCS is a core indicator in WFP programming and serves as the food-security benchmark for GFA in the DRC. It summarizes dietary diversity and consumption frequency over the previous seven days (WFP, 2008). FCS was collected for every census household, allowing us to evaluate coverage and targeting precision across the full study population.

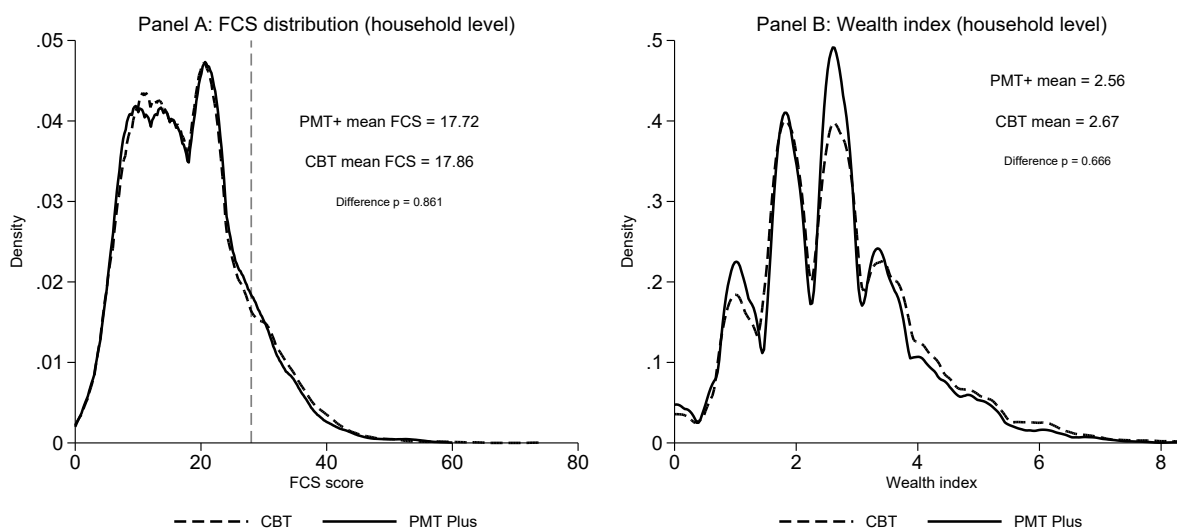
Using census FCS as the evaluation benchmark raises two concerns. First, it may mechanically favor PMT+, since the PMT component explicitly targets predicted FCS, whereas CBT committees may apply a different local definition of vulnerability. Therefore, as a robustness check, we construct a wealth index to assess whether our conclusions depend on using the outcome that PMT+ was designed to predict. The index is defined as the first principal component of eleven indicators of asset ownership, housing conditions, and access to services (Filmer and Pritchett, 2001). Its correlation with FCS is only 0.29, suggesting that the two measures capture different dimensions of vulnerability.

The second concern is whether eligibility should have been based directly on census FCS, which was observed for every household, rather than on the PMT model’s predicted FCS. Directly targeting on observed FCS, however, would not reflect operational practice. Financial and logistical constraints generally prevent agencies from collecting detailed food-consumption data for an entire population. WFP also advises against targeting on outcomes that are volatile and may be misreported when households know that their responses deter-

mine eligibility.¹² We therefore withheld observed FCS from the team developing the PMT models and used it only to evaluate coverage and targeting performance.

Figure 1 presents the baseline distributions of FCS and wealth by targeting arm. Panel A shows that food insecurity is widespread, with an average of 86% of households across blocs reporting an FCS of 28 or lower. The threshold lies within a densely populated part of the distribution rather than at a natural break. Both distributions are balanced across targeting arms ($p = 0.86$ for FCS and $p = 0.67$ for wealth).

Figure 1: Baseline distributions of FCS and wealth, by targeting arm



Notes: Kernel densities from the baseline census. Panel A plots FCS. The dashed vertical line marks the food-insecurity threshold of 28. Panel B plots the wealth index, defined as the first principal component of eleven indicators of asset ownership, housing conditions, and access to services and shifted to be nonnegative. The measures are balanced across targeting arms ($p = 0.86$ and $p = 0.67$) and weakly correlated ($\rho = 0.29$).

Administrative records: We compiled WFP’s administrative records of three types for our analysis. First, SCOPE, WFP’s beneficiary registration platform, identifies the households that received transfers and allows us to assess adherence to the targeting protocol.

¹²See WFP’s *Targeting Operational Guidance* about targeting using directly measured outcomes: <https://vamresources.manuals.wfp.org/docs/targeting-operational-guidance>.

Second, the country office’s Complaints and Feedback Mechanism records formal grievances submitted by excluded households. Third, program financial records allow us to estimate the cost of implementing each targeting approach.

Household survey data: Our main analysis draws on two rounds of survey data collected from the evaluation sample of 3,998 households.¹³ The first survey follow-up (FU1) reached 3,460 households (about 41 per bloc): 1,753 in PMT+ blocs (69% beneficiaries) and 1,707 in CBT blocs (64% beneficiaries). The second survey follow-up (FU2) reached 3,960 households (47 per bloc): 2,016 in PMT+ blocs (70% beneficiaries) and 1,944 in CBT blocs (64% beneficiaries). FU2 added modules on social cohesion, trust, and tension that we did not expect to move immediately. Both rounds were addressed to the female head of household, with a knowledgeable adult woman or, failing that, a same-status replacement household substituted when she was unavailable.

3.2 Outcome and Performance Measures

We evaluate the targeting methods along two dimensions: their precision in identifying food-insecure households and their downstream effects on households and communities. We measure targeting precision using the baseline census and downstream outcomes using the follow-up surveys and WFP administrative records.

Targeting performance: We construct each targeting indicator from the full census and aggregate it to the bloc level. We report the coverage rate, defined as the share of households selected for assistance in each bloc and averaged across blocs within each arm. We also report four precision measures that capture two dimensions of targeting performance: how accurately each method classifies households relative to the FCS cutoff and how well it ranks households across the full distribution of need. The exclusion and inclusion error rates

¹³The study was pre-registered with the AEA RCT Registry (AEARCTR-0010350), and its design received ethical approval from the University of Milan-Bicocca Institutional Review Board (approval no. 759); see Section E for consent and registration details.

capture the first dimension and are calculated following [Aiken et al. \(2022\)](#). The Coady–Grosh–Hoddinott index ([Coady et al., 2004](#)) and the Targeting Differential ([Galasso and Ravallion, 2005](#)) capture the second and are evaluated at the bottom 10%, 20%, and 40% of the FCS distribution. We report every measure against both benchmarks, the FCS and the wealth index described in Section 3.1. Appendix B provides the exact definitions.

Downstream outcomes: We measure household welfare using five outcomes. FCS captures the frequency and diversity of food consumption. The reduced Coping Strategies Index (rCSI) measures reliance on short-term responses to food shortages, such as reducing meal size or frequency ([Maxwell and Caldwell, 2008](#)). Livelihood coping captures more severe responses that may undermine future livelihoods, such as selling productive assets, and enters as three indicators for whether the household used any stress, crisis, or emergency strategy ([WFP, 2021](#)). A life-satisfaction score measures subjective well-being using three self-anchoring ladder questions ([Cantril, 1965](#)). Respondents place themselves on a ten-step staircase according to their current life satisfaction, the respect they receive within the community, and the extent to which their opinions are valued; we combine the three into an inverse-covariance-weighted index ([Anderson, 2008](#)). Finally, a women’s agency index, following [Jayachandran et al. \(2023\)](#), measures decision-making power and freedom of movement.

Beyond welfare, we examine three indices capturing households’ experiences of the targeting process and its broader social consequences. The satisfaction index combines overall satisfaction with perceptions of fairness, transparency, and timeliness, together with formal grievances recorded in WFP administrative data. The social cohesion index aggregates interpersonal and institutional trust, social closeness, and collective action. The community tensions index captures perceived safety, tensions within the community, and favoritism in beneficiary selection. These outcomes were collected at the second follow-up, allowing time for community-level effects to emerge. Appendix B defines each measure.

Index construction: The women’s-agency index, the satisfaction index, the social-cohesion index, and the tension index are each constructed as inverse-covariance-weighted averages of their standardized components, following [Anderson \(2008\)](#): each component is standardized to mean zero and unit standard deviation, the components are aggregated with weights given by the inverse of their covariance matrix, and the composite is re-standardized. Appendix Table [B1](#) lists each index and its component survey items.

3.3 Timeline

The census was conducted between December 2022 and March 2023, before any targeting decision or transfer. Cash transfers were then disbursed in four monthly instalments between May and September 2023, delivered at designated distribution points to households registered in SCOPE. The first follow-up survey (FU1) was fielded in September–October 2023, immediately after the final transfer, and the second (FU2) in May–June 2024, roughly eight months after the program ended.

Data collection took place in an active-conflict environment, and security shaped both timing and coverage. During FU1, severe disruptions affected about 800 households (20% of the planned sample), compounded by initial refusals by some local leaders to authorize interviews and the seasonal hunting season; replacement households were interviewed in 262 of these cases, and replacements accounted for 7.5% of FU1 and 8.2% of FU2 interviews, with no systematic difference across arms. By FU2, conditions had stabilized and nearly the full target sample was reached. In both rounds, enumerators worked in gender-balanced teams, were kept unaware of each bloc’s treatment assignment, and completed a one-week training followed by field piloting before data collection began.

4 Empirical Strategy and Internal Validity

4.1 Empirical Strategy

Our analysis proceeds at two levels that mirror the experimental design. We assess coverage and targeting precision at the bloc level, the unit of randomization, using the full baseline census. We measure downstream outcomes at the household level using the follow-up surveys, focusing on welfare and indices of satisfaction, social cohesion, and intergroup tensions.

Coverage and targeting performance: We begin by estimating the effect of the targeting method on coverage. We use the following bloc-level specification:

$$c_b = \alpha CBT_b + \eta_q + u_b, \tag{1}$$

where c_b is the share of census households selected in bloc b , CBT_b indicates assignment to community-based targeting, and η_q denotes randomization-stratum fixed effects. The coefficient α captures the average difference in coverage between CBT and PMT+. We report heteroskedasticity-robust standard errors for all bloc-level regressions, allowing residual variance to differ across census-based bloc aggregates.

Variation in coverage matters for our empirical analysis because it mechanically constrains exclusion error. A method covering a share c_b of a bloc in which a share π_b is in need must exclude at least $1 - c_b/\pi_b$ of households in need, even under perfect ranking. For example, if 80% of households are in need, the minimum exclusion error is 0.14 when coverage is 0.69, but zero when coverage is 0.80. Coverage-free error rates therefore conflate ranking quality with caseload size. To account for variation in coverage, we present targeting performance at realized coverage alongside two comparisons that account for differences in caseload size. The first constructs an equal-coverage allocation, while the second controls for coverage in the regression.

At realized coverage, we estimate

$$y_b = \beta CBT_b + \eta_q + \varepsilon_b, \quad (2)$$

where y_b is a targeting-performance measure in bloc b and β captures the difference between CBT and PMT+ at the caseloads each method produced. We estimate equation (2) for exclusion errors, inclusion errors, the Coady–Grosh–Hoddinott index (Coady et al., 2004), and the Targeting Differential (Galasso and Ravallion, 2005). We repeat the analysis using the wealth index as a robustness check.

For the equal-coverage comparison, we apply the PMT+ selection rule within each CBT bloc, ranking households by their PMT+ score and selecting the same number as the committee. We adjust the threshold $\tilde{\tau}_b$ so that the selected share equals realized CBT coverage, c_b^{CBT} :

$$\frac{1}{N_b} \sum_{j \in b} \mathbb{1}\{s_j^{\text{PMT}} \geq \tilde{\tau}_b\} = c_b^{\text{CBT}}, \quad (3)$$

where N_b is the number of households in bloc b , with ties at the threshold resolved deterministically to match the caseload exactly. We then compare the observed CBT and simulated PMT+ allocations using

$$y_{bm} = \gamma CBT_{bm} + \eta_q + \varepsilon_{bm}, \quad (4)$$

where m indexes the two allocations and CBT_{bm} equals one for CBT and zero for simulated PMT+. Since the allocations have identical caseloads within each bloc, γ captures differences in targeting performance arising from which households are selected rather than how many receive assistance.

The coverage-controlled comparison instead adds the bloc coverage rate c_b to equation (2):

$$y_b = \beta_c CBT_b + \delta c_b + \eta_q + \varepsilon_b. \quad (5)$$

This specification compares observed performance across arms conditional on coverage, with-

out constructing a simulated allocation. This specification is econometrically more transparent, since it retains the experimental contrast between arms rather than relying on a simulated counterfactual. However, it carries its own cost. Coverage is itself an outcome of the targeting method, so conditioning on it compares blocs that reached similar caseloads through different underlying need. The two comparisons are therefore complements. The simulation holds caseload fixed by construction but is deterministic, while the coverage control preserves the randomization but conditions on a post-treatment variable. We report both.

Downstream household-level outcomes: For welfare and other downstream outcomes such as satisfaction and social cohesion estimate intention-to-treat effects across the two follow-up rounds,

$$y_{ib} = \tau \text{ CBT}_b + Z_i' \theta + \eta_q + \epsilon_{ib}, \quad (6)$$

where y_{ib} is an outcome for household i in bloc b —the FCS, the reduced Coping Strategies Index, satisfaction with targeting, or trust, for example—and τ is the effect of residing in a CBT bloc. The vector Z_i holds baseline census controls chosen by double-LASSO ([Urminsky et al., 2016](#)), always including the baseline value of the outcome as a non-penalized regressor, and standard errors are clustered at the bloc level. The survey sampled equal numbers of eligible and ineligible households while roughly four-fifths of the study population is eligible, so ineligible households are over-represented relative to the bloc census. Because treatment is randomized within quadruplets, the unweighted estimator identifies the intention-to-treat effect consistently ([Solon et al., 2015](#)), and unweighted estimates are our primary specification. As a robustness check, and to recover population-representative levels, we reweight to the census with a calibrated population weight—the sampling design weight for all sampled households, including replacements, raked to census margins on the design cell and eleven household characteristics and winsorized at the 99th percentile.

Since coverage differs across arms, τ combines the effect of the selection method with

that of serving fewer households. We therefore also estimate (6) adding the bloc coverage rate, so that the CBT coefficient reflects differences conditional on caseload. We examine heterogeneity in the same way as for precision, interacting CBT_b with an indicator for blocs above the median in bloc size, realized coverage, or the share of food-insecure households, with bloc-clustered standard errors throughout.

4.2 Internal Validity

Randomization produced broadly comparable treatment arms. Appendix Table A3 reports balance across a set of baseline characteristics. The variables most relevant to targeting are closely matched. Mean FCS is 17.8 in PMT+ blocs and 18.7 in CBT blocs, the share of food-insecure households is 88% and 86%, respectively, and the mean wealth index is 2.79 and 2.83. A small number of covariates exhibit modest differences. Household heads in CBT blocs are, on average, 0.8 years older, and crisis and emergency coping strategies are 4 and 7 p.p. more common, respectively. Among the 47 baseline covariates measured in the census, 7, 4, and 3 are statistically significant at the 10, 5, and 1 percent levels, respectively. Random assignment remains the basis for identification. In all specifications, we use double LASSO to select prognostic baseline controls, which improves precision and accounts for any residual imbalance.

Attrition was moderate and balanced across treatment arms. Appendix Table A4 reports attrition and replacement rates by survey round and treatment arm. At the first follow-up, approximately 20% of target households could not initially be reached because of insecurity in specific blocs, refusals by some local leaders to authorize interviews, and the seasonal hunting period. After replacements, the realized sample included 3,460 households, corresponding to an attrition rate of 13.5%. Conditions improved by the second follow-up. Attrition fell to 9% before replacements and 1% afterward, yielding a realized sample of 3,960 households. Replacement households accounted for 7.5% and 8.2% of interviews in the first and second follow-ups. Differences in attrition between the treatment arms are small and statistically

insignificant in both rounds.

Implementation closely followed the experimental assignment in both arms. Appendix Table A5 reports transfer receipt using WFP’s SCOPE administrative records. Among eligible households, 93% received at least one transfer and 82% received all four. The corresponding rates are 94% and 84% under PMT+ and 92% and 81% under CBT. Only one of the 1,294 ineligible households received any assistance.

The balance, attrition, and implementation checks support the internal validity of the experiment. One remaining limitation is the security-driven exclusion of 12 blocs, which reduced statistical power and may limit the external validity of the findings. Because these blocs were excluded before randomization and the baseline census, their exclusion does not compromise the randomized comparison among the 84 blocs retained in the study. As specified in the study design, we also re-estimate the main results after excluding all randomization strata affected (quadruplets) by the loss of a bloc. The findings remain unchanged.

5 Results

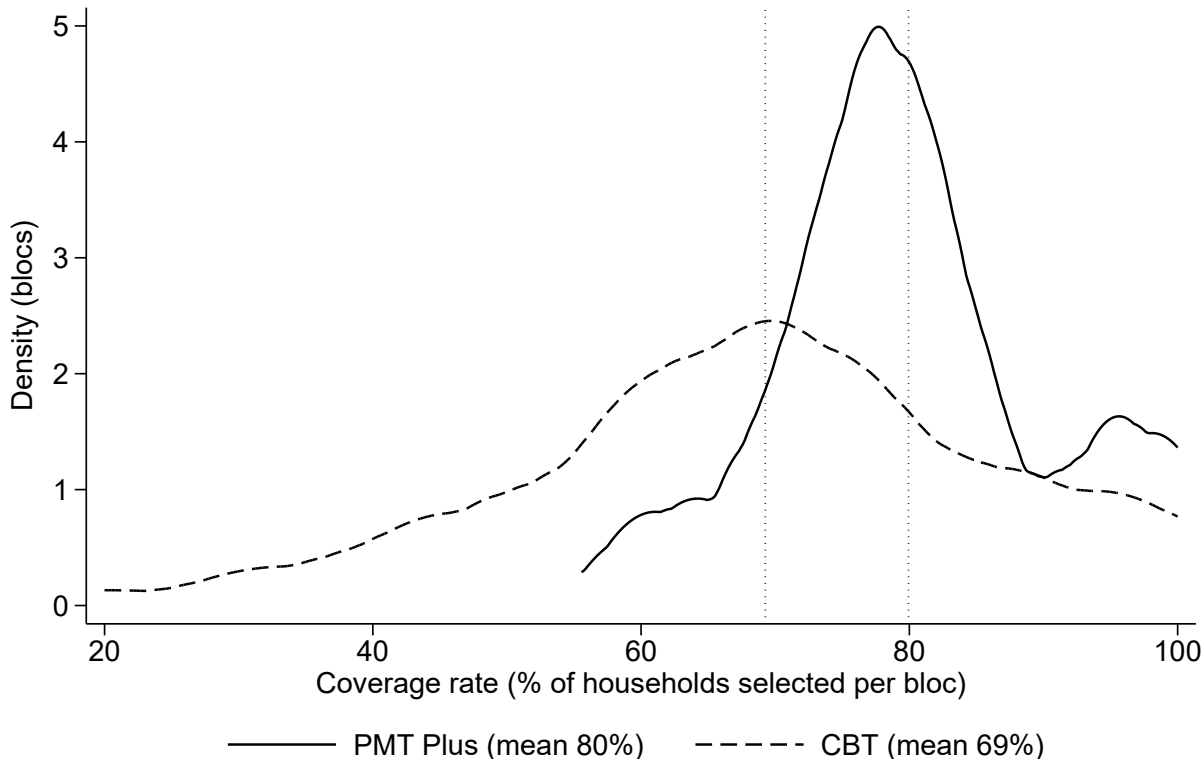
5.1 Coverage and Eligibility

The first outcome we examine is coverage. Since neither targeting method was subject to a quota and every household deemed eligible received assistance, each method directly determined its own caseload. PMT+ selected 80% of households, compared with 69% under CBT. This 11 p.p. difference is operationally large and statistically significant ($p < 0.01$).¹⁴ Figure 2 further shows that coverage varied more across CBT blocs. Notably, under both methods, the share of households selected falls below the 86% classified as food insecure in the census. The gap between coverage and measured food insecurity may reflect overreporting in the census, imperfect assessments of need, including prediction error under PMT+, or

¹⁴Across the 42 CBT blocs, which contain 5,923 census households, an 11 p.p. coverage difference corresponds to approximately 650 households, or 15 per bloc. At the program’s transfer value of \$774 per household over four distributions, this difference represents approximately \$0.5 million in transfers.

both. Our data, nonetheless, do not allow us to disentangle these explanations.

Figure 2: Coverage by targeting method



Notes: Kernel densities of the share of census households selected in each bloc, with 42 blocs per arm. Solid and dashed lines represent PMT+ and CBT, respectively. Dotted lines mark arm means (79.9% and 69.3%). The CBT effect is -10.8 percentage points (SE 3.6, $p < 0.01$), estimated with quadruplet fixed effects and bloc-clustered standard errors. Coverage standard deviations are 10.5 and 18.2 percentage points under PMT+ and CBT, respectively.

Despite facing no quota, community committees selected fewer beneficiaries than PMT+, contrary to concerns that local discretion may broaden eligibility. This coverage margin is absent from most comparisons, which either hold caseloads fixed across targeting methods (Alatas et al., 2012; Hillebrecht et al., 2020; Premand and Schnitzer, 2021) or allow communities to choose between broader coverage and larger transfers (Abay et al., 2026). The implications of lower coverage are ambiguous, since selecting fewer households within each community may allow a fixed budget to reach more communities when needs exceed available

resources, but may also reduce targeting precision and welfare or generate additional social costs. We examine these tradeoffs next.

Eligibility

The two methods differ not only in caseload size but also in whom they select. CBT and PMT+ rank households using different information and therefore prioritize different dimensions of vulnerability. Appendix Table A6 reports the ten largest statistically significant differences in baseline characteristics between households selected under CBT and those PMT+ would select at the same coverage, using the equal-coverage simulation within CBT blocs. Differences are ordered by magnitude, and characteristics with no significant difference are omitted. Relative to PMT+, CBT selects 4.6 p.p. more agriculture-dependent households, 3.4 p.p. more households with low production, 3.0 p.p. more households with a pregnant or lactating woman, and 2.4 p.p. more households that recently lost crops to pests. PMT+ instead selects more households headed by an illiterate person (5.2 p.p.), a displaced person (2.1 p.p.), or a woman (1.2 p.p.). The two methods thus reflect different but defensible notions of vulnerability. CBT places greater weight on current agricultural and seasonal risks, whereas PMT+ places greater weight on persistent demographic characteristics.

Since coverage falls below measured need, we use the relative-representation index (RR) to assess whether particular groups are overrepresented or underrepresented relative to their share of food-insecure households (Appendix Table A7).¹⁵ Most groups are underrepresented under both methods. CBT gives greater representation to agriculture-dependent households (RR 1.02 versus 0.95, $p < 0.10$) and those with a pregnant or lactating woman (1.20 versus 0.95). PMT+ more strongly overrepresents households with an illiterate head (1.18 versus 1.02, $p < 0.01$). Two households are similarly represented across methods. Displaced households are substantially underrepresented under both, particularly CBT (RR 0.26 versus 0.44

¹⁵The index divides a group's share of beneficiaries by its share of food-insecure households, defined as census FCS of 28 or lower. Values below one indicate underrepresentation and values above one indicate overrepresentation.

under PMT+).

These profile differences suggest that CBT gives weight to dimensions of vulnerability not fully captured by the PMT score. Appendix Table A8 examines this possibility by relating CBT selection to recent shocks and vulnerability indicators, conditional on predicted FCS. Holding the PMT score constant, committees are 11 p.p. more likely to select households with a chronically ill head. The corresponding differences for households headed by a person with a disability and those with a displaced member are both 26 p.p. These associations are consistent with committees drawing on additional local information.

5.2 Targeting Precision

Table 1 reports precision under three comparisons. The raw comparison evaluates PMT+ and CBT at their realized caseloads. The equal-coverage comparison applies a simulated PMT+ rule within each CBT bloc, matching that bloc’s realized coverage. The coverage-controlled comparison adds realized bloc coverage to the regression across randomized arms.

At realized coverage, inclusion errors are similar across methods: 11% of households selected under PMT+ and 13% under CBT are not food insecure according to our FCS benchmark. This 2 p.p. difference is not statistically significant. Exclusion errors differ more substantially. CBT leaves 29% of food-insecure households unselected, compared with 19% under PMT+, an estimated difference of approximately 11 p.p. that is significant at the 1% level.

Both coverage adjustments substantially reduce this gap. At CBT’s realized coverage, the simulated PMT+ rule excludes 28.5% of food-insecure households, compared with 29.0% under CBT, an insignificant difference of 0.5 p.p. Adding realized coverage to the regression yields a similarly small and insignificant difference of 0.9 p.p. Inclusion errors remain statistically indistinguishable under both adjustments. These results suggest that PMT+’s lower raw exclusion error reflects its larger caseload rather than more accurate household rankings.

The distribution-sensitive measures yield similar results. Neither the aggregate CGH index nor the Targeting Differential differs significantly across methods under any comparison. Some tail estimates differ in the raw and coverage-controlled comparisons, but these differences are sensitive to how ties in the underlying scores are resolved and disappear at equal coverage.¹⁶ We therefore find no robust evidence that either method reaches the most vulnerable households more effectively.

As a robustness check, Appendix Table A9 repeats the analysis using the wealth index.¹⁷ The results mirror those for food insecurity. At realized coverage, CBT excludes more asset-poor households than PMT+ (27.3% versus 18.2%, $p < 0.05$). At equal coverage, exclusion errors are nearly identical at 27.4% under simulated PMT+ and 27.3% under CBT. Neither the aggregate CGH index nor the Targeting Differential differs significantly across methods, and no tail differences are significant under the reported tie-breaking rule. The wealth results thus reinforce the finding of similar targeting precision at equal coverage.

¹⁶Percentile cutoffs often fall within groups of households with identical scores, making tail estimates sensitive to tie-breaking rules.

¹⁷Since the wealth index has no natural eligibility threshold, we classify the bottom 74.6% of households by wealth rank as asset poor. This share equals average realized coverage across the two arms, the midpoint of 79.9% under PMT+ and 69.3% under CBT.

Table 1: Targeting precision under the food-security benchmark

	Coverage free		Equal coverage		Coverage controlled	
	PMT+	CBT	PMT+	CBT	PMT+	CBT
Exclusion error	0.189	0.290*** (0.036)	0.285	0.290 (0.006)	0.235	0.244 (0.010)
Inclusion error	0.110	0.129 (0.020)	0.122	0.129 (0.007)	0.112	0.126 (0.023)
CGH index	1.012	1.030 (0.027)	1.038	1.030 (0.008)	1.027	1.015 (0.018)
CGH (bottom 10%)	1.041	0.925* (0.079)	0.939	0.925 (0.056)	1.049	0.917 (0.084)
CGH (bottom 20%)	1.059	1.009 (0.070)	1.035	1.009 (0.035)	1.078	0.991 (0.064)
CGH (bottom 40%)	1.038	1.012 (0.051)	1.032	1.012 (0.028)	1.061	0.989* (0.040)
Targeting Differential	0.079	0.041 (0.054)	0.068	0.041 (0.033)	0.101	0.019 (0.052)
TD (bottom 10%)	0.052	-0.006 (0.042)	-0.006	-0.006 (0.048)	0.057	-0.011 (0.046)
TD (bottom 20%)	0.075	0.031 (0.036)	0.045	0.031 (0.032)	0.087	0.019* (0.037)
TD (bottom 40%)	0.061	0.012 (0.039)	0.035	0.012 (0.029)	0.080	-0.007** (0.035)
Number of blocs	84		42		84	

Notes: The table reports bloc-level targeting performance against the census food-security benchmark (FCS of 28 or lower). Raw columns evaluate each arm at its realized caseload. Equal-coverage columns compare CBT with a simulated PMT+ allocation matched to coverage within each CBT bloc. Coverage-controlled columns add realized bloc coverage to the regression. All regressions include quadruplet fixed effects. Bloc-clustered standard errors are in parentheses. Stars indicate significance of the CBT–PMT+ difference: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Our error rates are relatively low, but comparisons with the targeting literature must account for differences in coverage and error definitions. Table C1 compares studies reporting errors for both data-driven and community-based methods; the discussion accompanying that table gives the definitions and limitations. Exclusion errors generally decline as coverage expands, since fewer households in need remain unserved. Studies with coverage comparable to ours report errors ranging from 0.11 to 0.14 in Malawi at 80% coverage (Basurto et al., 2020) and from 0.17 to 0.47 in Cameroon at 67% coverage (Stoeffler et al., 2016). The higher exclusion errors elsewhere in the table, ranging from 0.36 to 0.72, come from programs reaching only 20 to 45% of households. These comparisons provide context rather than a ranking of targeting performance. Our low inclusion errors also partly reflect widespread food insecurity, which limits the share of households that can be incorrectly included.

The similar precision of PMT+ and CBT differs from evidence that PMT more accurately identifies consumption-poor households (Alatas et al., 2012; Premand and Schnitzer,

2021; Schnitzer and Stoeffler, 2024). One possible explanation is the benchmark. Models based on slow-moving household characteristics may capture structural poverty but miss shocks and seasonal variation (Del Ninno and Mills, eds, 2015). Their advantage may therefore weaken when performance is assessed against acute food insecurity rather than chronic poverty, consistent with Premand and Schnitzer (2021), who find similar performance between community-based targeting and FCS-based PMT in Niger.

Heterogeneities: We do not find evidence that the similar performance of the two methods mask heterogeneity across communities. Appendix Table A10 examines whether relative targeting performance varies with the prevalence of food insecurity and within-bloc inequality, measured by the FCS coefficient of variation. The exclusion-error gap between CBT and PMT+ increases with prevalence ($\beta = 0.70$, $p < 0.10$), but the coefficient falls to 0.23 and becomes statistically insignificant after controlling for coverage. We find no corresponding heterogeneity in inclusion errors, nor does either error gap vary significantly with FCS dispersion.

5.3 Downstream Welfare Outcomes

Table 2 reports intention-to-treat effects of CBT relative to PMT+ across the full study population, including beneficiaries and nonbeneficiaries, at both follow-ups. Despite covering 11 percentage points fewer households, CBT generates similar average food-security and well-being outcomes. Its estimated effects on FCS are 0.22 points at the first follow-up and 0.27 at the second, relative to PMT+ means of 18.1 and 14.6, with neither estimate statistically significant. Food- and livelihood-based coping strategies and life satisfaction likewise show no meaningful differences. Women’s agency estimates are small and consistently positive, but their significance varies across specifications, providing no robust evidence of a difference between methods. The conclusions are unchanged when the sample is reweighted to the bloc census (Appendix Table A11). These null effects are not driven by low statistical power. The minimum detectable effects are 0.18 of a control-group standard deviation for FCS at the

first follow-up and 0.09 at the second, with comparable values for the remaining outcomes (Appendix Table A12). We can therefore rule out differences larger than a small fraction of a standard deviation.

Table 2: Effect of CBT on household welfare (intention-to-treat)

	Follow-up 1		Follow-up 2	
	PMT+ mean (1)	CBT effect (2)	PMT+ mean (3)	CBT effect (4)
Food Consumption Score	18.118	0.222 (0.625)	14.561	0.271 (0.340)
Reduced Coping Strategies Index (rCSI)	21.182	0.434 (0.553)	25.198	0.092 (0.555)
Used stress coping strategy	0.966	-0.013 (0.009)	0.952	-0.001 (0.006)
Used crisis coping strategy	0.754	-0.033 (0.021)	0.774	0.014 (0.014)
Used emergency coping strategy	0.618	-0.042 (0.036)	0.711	-0.028 (0.020)
Life satisfaction (Anderson index)	-0.031	0.053 (0.055)	0.004	-0.023 (0.040)
Women’s agency (Anderson index)	-0.029	0.058 (0.037)	-0.010	0.017 (0.032)

Notes: Each row reports the ITT effect of assignment to CBT relative to PMT+ on the outcome, from equation (6): an unweighted household-level regression with quadruplet fixed effects, double-LASSO-selected baseline controls (including the baseline outcome), and standard errors clustered at the bloc level. Columns (1) and (3) report the PMT+ mean at the first and second follow-up; columns (2) and (4) report the CBT effect with its standard error below. The conclusions are unchanged when the sample is reweighted to the bloc census with the census-calibrated population weight. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Heterogeneities: We examine whether the similar average welfare outcomes across methods mask heterogeneity by bloc size, coverage rate, and baseline food insecurity (Appendix Table A13). A consistent pattern emerges in larger blocs, where CBT reduces FCS by 2.5 points at the first follow-up and 1.5 at the second and increases reliance on consumption-based coping. One possible explanation is that committees know less about individual households in larger communities, making it harder to identify those most in need. Our precision results above, however, provide no evidence that CBT performs worse in larger communities.

Reliance on coping strategies is also higher in high-coverage blocs at the second follow-up, in both the unweighted and weighted specifications, though the pattern does not carry to the first round or to the other outcomes. The remaining estimates show no consistent pattern across outcomes or survey rounds, and with 48 interactions estimated we interpret these isolated differences cautiously.

The transfer's own effect

One possible explanation for the absence of welfare differences between targeting methods is that the assistance itself had no effect. Although our experiment identifies the effect of the targeting method rather than the transfer, we assess this possibility using a difference-in-differences comparison of beneficiaries and nonbeneficiaries across survey rounds, exploiting the panel structure of the data.¹⁸ Appendix Table A14 presents this comparison for FCS, the only outcome measured in both the baseline census and the two follow-up surveys. The coefficient captures the change in FCS among beneficiaries relative to the corresponding change among nonbeneficiaries between baseline and each follow-up. Assistance leads to an increase in beneficiaries' FCS of 3.0 points at the first follow-up and 1.9 points at the second, with both estimates significant at the 1% level.

Welfare consequences of lower coverage

CBT's smaller caseload could, in principle, harm the households it leaves out. Adding bloc coverage as a control leaves the welfare estimates substantively unchanged (Appendix Table A15), and the comparison among non-beneficiaries provides no consistent evidence of worse outcomes under CBT (Table 3). Food security, coping strategies, and life satisfaction are similar across arms. Women's agency estimates are small and positive, but only marginally significant in two specifications and not robust across specifications. Non-

¹⁸While beneficiary status was not randomly assigned and we have only one pre-intervention observation, preventing a direct test of the parallel-trends assumption, random assignment of the targeting method places households within a common experimental and implementation environment across arms. This limits the scope for differences in program conditions to explain the estimates. Nonetheless, we interpret these results as suggestive.

beneficiaries may, however, differ across arms because each method selects different households. We therefore also compare outcomes among all households classified as food insecure at baseline, regardless of whether they subsequently received assistance. This comparison uses a group defined before targeting and likewise shows no significant differences in food security, coping strategies, or life satisfaction (Appendix Table A16).

Table 3: Effect of CBT on non-beneficiary welfare

	PMT+ non-benef. mean (1)	CBT effect (Follow-up 1) (2)	CBT effect (Follow-up 2) (3)
Food Consumption Score	14.536	1.615* (0.865)	0.293 (0.418)
Reduced Coping Strategies Index (rCSI)	24.902	0.298 (0.792)	1.241** (0.578)
Used stress coping strategy	0.965	-0.006 (0.011)	-0.000 (0.012)
Used crisis coping strategy	0.773	-0.027 (0.035)	0.017 (0.026)
Used emergency coping strategy	0.718	-0.042 (0.046)	-0.044 (0.035)
Life satisfaction (Anderson index)	0.061	0.010 (0.080)	-0.062 (0.053)
Women’s agency (Anderson index)	-0.055	0.119* (0.066)	0.054 (0.048)

Notes: Intention-to-treat effect of CBT relative to PMT+ estimated on non-beneficiary (unselected) households only, from equation (6): unweighted, quadruplet fixed effects, double-LASSO controls, bloc-clustered standard errors. Column (1) is the PMT+ non-beneficiary mean; columns (2) and (3) report the CBT effect at each follow-up. The absence of any systematic harm to non-beneficiaries is unchanged when the sample is reweighted to the bloc census with the census-calibrated population weight, with the handful of marginally significant cells shifting under weighting. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.4 Satisfaction, Social Cohesion, and Other Trade-offs

Targeting methods may affect both perceptions of beneficiary selection and relations within communities. Table 4 reports ITT estimates of CBT’s effects on indices of satisfaction with the targeting process, social cohesion, and tensions, alongside specifications accounting for differences in coverage and population reweighting. These outcomes are measured at the

second follow-up to allow time for effects to emerge.

The satisfaction index does not differ significantly across arms in the primary specification (-0.08 , standard error 0.05). The estimate becomes marginally significant after controlling for coverage (-0.09 , $p < 0.10$), but is not robust to reweighting. Component-level differences are limited to fairness and timeliness: households in CBT blocs are less likely to describe the process as fair (-0.040 , $p < 0.10$) and more likely to describe it as slow (0.059 , $p < 0.01$), consistent with the additional time required by committee selection. WFP’s administrative complaints records complement these self-reported measures. Of 651 cases recorded across the 84 study blocs, 148 concern targeting, including 50 alleged inclusion or exclusion errors and 98 requests for assistance following non-selection. The remaining 503 concern collection procedures. Neither the presence of targeting complaints in a bloc nor their number or rate per hundred households differs significantly across arms (See Appendix Table [A18](#) for details).

The social cohesion and tension indices likewise show no significant differences across arms. Among the tension components, the primary estimates for feeling unsafe at home (0.040) and in the community (0.041 , $p < 0.10$) provide limited evidence of differences across arms. Both become larger and significant at the 5% level or below after reweighting (Appendix Table [A17](#)), but our data do not allow us to identify the reasons for these feelings of insecurity.

Table 4: Effect of CBT on satisfaction, social cohesion, and tension

	PMT+ mean (1)	CBT effect		
		Unadjusted (2)	Coverage-adjusted (3)	Weighted (4)
Satisfaction index	0.037	-0.077 (0.048)	-0.093* (0.054)	-0.036 (0.055)
Social cohesion index	-0.008	0.020 (0.041)	-0.038 (0.038)	-0.101 (0.080)
Tension index	-0.011	0.019 (0.050)	-0.064 (0.045)	0.177 (0.116)

Notes: Outcomes are inverse-covariance-weighted indices of standardized components measured at the second follow-up. Satisfaction captures overall satisfaction, fairness, transparency, and timeliness. Social cohesion captures mutual support, collective action, and trust. Tensions capture perceived insecurity and recent conflict, with higher values indicating more tension. Column (1) reports the PMT+ mean. Columns (2)–(4) report CBT effects from the primary, coverage-controlled, and population-reweighted specifications. All include quadruplet fixed effects and double-LASSO-selected baseline controls. Bloc-clustered standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A common concern with community-based targeting is favoritism toward households connected to the selection committee. Within CBT blocs, neither having a household member on the committee nor knowing a committee member is significantly associated with selection, with or without controls for need. Households in CBT blocs are marginally more likely to report that some households have better connections, relative to a PMT+ mean of 2.5%, but we find no corresponding association between reported connections and beneficiary selection. Appendix Table A19 reports both sets of estimates.¹⁹

6 The Cost of Targeting

The preceding sections show that PMT+ and CBT achieve similar targeting precision at equal coverage and comparable average welfare outcomes, despite substantially lower coverage under CBT. These findings raise two questions under constrained humanitarian budgets:

¹⁹Connections are self-reported at the second follow-up and subject to substantial item nonresponse. Beneficiary status may also affect reported awareness of the committee. These estimates are therefore suggestive rather than causal.

whether targeting is worthwhile when need is widespread, and which method delivers comparable outcomes at lower cost. Given the absence of detectable welfare differences, we assess whether the gains in targeting performance justify the costs of screening. We then use a cost-minimization simulation to examine how CBT’s lower coverage, and hence lower transfer costs per community, could allow a fixed budget to reach more communities while maintaining comparable average welfare outcomes.

Table 5 reports targeting and transfer costs. Set-up and screening costs are nearly identical across arms, at approximately USD 61,900 for PMT+ and USD 62,600 for CBT, or USD 10.65 and USD 10.56 per household screened, respectively. Both methods used the same census for experimental purposes, although CBT could potentially rely on a shorter questionnaire or less frequent data collection in practice. Screening costs are small relative to the household transfer, at about 1.4% of its USD 774 value. Differences in total costs arise primarily from coverage, as PMT+’s larger caseload raises transfer expenditure and the cost of serving each bloc.

Table 5: Cost-effectiveness and budget reach by targeting method

	PMT+	CBT
<i>Program costs</i>		
Targeting / set-up cost, total (USD)	61,880	62,570
Screening cost per screened household (USD)	10.65	10.56
Households reached	4,930	4,396
Beneficiaries reached (individuals)	29,369	26,670
Transfer per assisted household (USD)	774	788
Total transfers (USD)	3,817,970	3,463,433
Total cost (USD)	3,879,850	3,526,002
<i>Budget reach ($G_m = B/C_m$)</i>		
Cost per bloc C_m (USD)	87,095	78,443
Blocs served at $B = \text{USD } 2\text{m}$	23	25
Blocs served at $B = \text{USD } 10\text{m}$	115	127

Notes: Set-up/targeting and screening costs are from program cost records; transfers are computed as the per-beneficiary transfer times the number of beneficiaries reached under each method. The budget-reach block reports the cost of serving one bloc, C_m , and the number of blocs a fixed budget B supports, $G_m = B/C_m$. These are program accounting records, not survey estimates, so no sampling weight applies.

What does screening buy at this level of need?

We first compare targeting with universal assistance within a community. Universal assistance eliminates exclusion errors but also reaches households that are not food insecure. Targeting reduces such inclusion at the cost of leaving some households in need unserved and incurring screening costs. We summarize this trade-off using

$$L_m = \lambda E_m \pi + (1 - \lambda) I_m c_m + \frac{s_m}{t}, \quad (7)$$

where E_m and I_m are the exclusion and inclusion error rates, π is the prevalence of food insecurity, c_m is coverage, s_m is the screening cost per household, and t is the household transfer. The parameter λ determines the relative weight placed on exclusion and inclusion errors.²⁰

Equating each method’s loss with that of universal assistance gives a prevalence threshold π^* above which universal assistance yields the lower calculated loss. Within a community with a secured budget (Table A20, Rule 1), the threshold falls below the observed mean prevalence of 0.86 at every weight considered. At $\lambda = 0.5$, it is 0.686 for PMT+ and 0.566 for CBT, falling to 0.071 and below zero, respectively, at $\lambda = 0.8$. Universal assistance yields a lower calculated loss in 95–99% of study blocs. At the observed mean prevalence and $\lambda = 0.8$, the loss is 0.03 under universal assistance, compared with 0.19 under PMT+ and 0.27 under CBT.²¹

We next assess the value of screening by comparing each method with random allocation at the same coverage (Table A21). PMT+ excludes 18.9% of food-insecure households, compared with 20.1% under random allocation. The corresponding rates for CBT are 29.0% and 30.7%, with similarly small reductions in inclusion errors. Net of screening costs, the

²⁰The weight λ reflects the planner’s preferences and is not identified by the experiment. We report results for $\lambda \in \{0.5, 0.7, 0.8, 0.9\}$. At $\lambda = 0.5$, the two errors receive equal weight. Our focal value of $\lambda = 0.8$ assigns four times as much weight to excluding a food-insecure household as to including one that is not.

²¹Each method’s arm-average error rates are applied to the full distribution of bloc prevalence, since each bloc is assigned to only one method. These calculations therefore hold error rates constant as prevalence varies.

calculated value of information at $\lambda = 0.8$ is -0.001 for PMT+ and 0.005 for CBT. Neither method yields a lower calculated loss than random allocation in more than roughly one-third of blocs. Under this benchmark and loss function, screening therefore provides limited gains over random selection at a fixed caseload. However, this comparison is based on the targeting precision and does not identify the welfare effects of random allocation.

These comparisons raise questions about the value of household targeting when need is widespread. Needs nevertheless far exceed available funding, and donor expectations that assistance prioritize the most vulnerable may limit the scope for universal provision. Identifying communities where universal assistance would be appropriate also requires precise estimates of need, which generally depend on household surveys. These resource, information, and accountability constraints suggest that targeting will remain central to humanitarian assistance (Banerjee et al., 2024).

New data sources and predictive methods may help address the information constraint. Machine learning applied to satellite imagery and other geospatial data can estimate local welfare without a census (Jean et al., 2016), and the resulting poverty maps can improve geographic targeting relative to survey-based allocation (Smythe and Blumenstock, 2022). Related methods also forecast acute malnutrition at operationally useful horizons (Constenla-Villoslada et al., 2025). These advances may offer less costly ways to identify communities where widespread need makes universal assistance appropriate.

The limited gains from screening offer one possible explanation for the similar welfare outcomes in Section 5. When food insecurity is widespread, methods that select different households may nevertheless reach populations with comparable needs. Our results therefore motivate examining whether CBT’s lower coverage and associated costs could allow assistance to reach more communities under a fixed budget without compromising average welfare.

Cost minimization and geographic reach

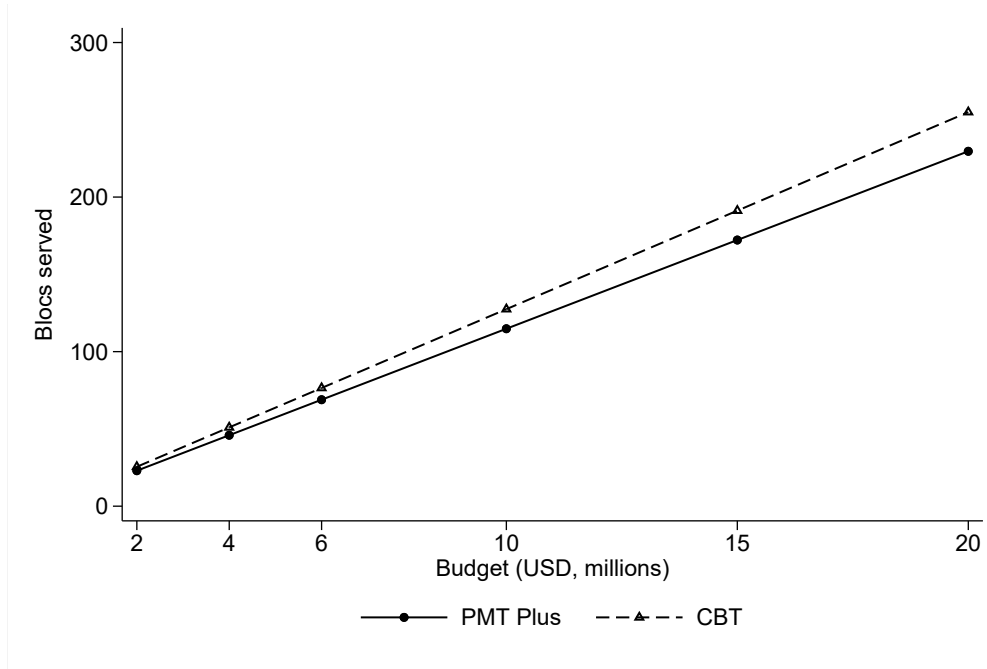
We assess now how differences in coverage and costs translate into the number of communities that can receive assistance under a fixed budget. For method m , the cost of serving an average bloc is

$$C_m = t k_m \bar{n} + s_m \bar{n}, \quad (8)$$

where t is the transfer per beneficiary household, k_m is the coverage rate, $\bar{n}$ is average bloc size, and s_m is the screening cost per household. Screening costs apply to all households, as both methods screen the full bloc. A fixed budget B therefore supports $G_m = B/C_m$ blocs.

Given the comparable average welfare outcomes, CBT's lower cost offers scope to extend geographic coverage. Serving one bloc costs approximately USD 87,100 under PMT+ and USD 78,400 under CBT. A USD 10 million budget would therefore reach about 115 blocs under PMT+ and 127 under CBT, an increase of roughly 11%. Figure 3 shows that this difference persists across budget levels, with the absolute gap growing as the budget increases. Broader geographic coverage does not necessarily mean reaching more households in need. The number reached per dollar is nearly identical across methods and under random allocation (Table A21). Rather, CBT allows assistance to extend to communities that would otherwise remain unserved, a potentially important margin when geographic equity and program presence matter alongside household-level targeting.

Figure 3: Number of blocs served under a fixed budget, by targeting method



Notes: CBT costs less per bloc, so a given budget reaches more communities under CBT than under PMT+, and the gap widens with the budget.

The role of coverage in generating these savings also suggests scope to refine CBT's design. Committees' assessments of need need not be fixed, and clearer eligibility guidance or safeguards for groups less well represented under CBT, such as displaced households, could reduce exclusion. Such changes would need to account for potential community tensions and the additional transfer costs of broader coverage. Our results therefore support considering CBT as an adaptable alternative to the survey-based PMT+ used in this experiment, rather than as a uniformly cheaper targeting technology.

7 Conclusion

This paper compares PMT+ and community-based targeting in a cluster-randomized experiment in Tanganyika Province, DRC. Unlike designs that fix beneficiary quotas, ours allows each method to determine its own caseload. Community committees selected 69% of house-

holds, compared with 80% under PMT+, despite baseline food insecurity affecting 86% of households. At equal coverage, the methods achieve similar targeting precision under both the food-security and wealth benchmarks. Average food consumption, coping strategies, and life satisfaction are also comparable across arms at both follow-ups, despite CBT's smaller caseload.

These findings suggest that the relative performance of targeting methods depends on both the objective and the coverage each method generates. Committees place greater weight on some dimensions of vulnerability not fully captured by the PMT score, potentially drawing on local information about household circumstances. Our data do not establish that this information explains the similar welfare outcomes, but they show that different selection rules can produce comparable precision when targeting acute food insecurity. Allowing coverage to vary also reveals that local discretion need not broaden eligibility. In this setting, committees applied a narrower definition of need than the data-driven rule.

The social results provide no consistent evidence of differences in satisfaction, social cohesion, or tensions at the index level, and reported committee connections do not predict selection. Some components nevertheless warrant attention. Households in CBT blocs perceive selection as slower and somewhat less fair, while evidence of greater perceived insecurity is sensitive to weighting. These findings suggest that comparable average welfare need not imply an identical experience of targeting.

For agencies facing constrained budgets, the cost implications arise primarily from coverage rather than screening. Both methods used the same census and incurred similar targeting costs, but CBT's smaller caseload reduced transfer expenditure per community. At the estimated costs, a USD 10 million budget could serve approximately 127 blocs under CBT, compared with 115 under PMT+. This gain concerns geographic reach rather than the number of food-insecure households reached per dollar, which is similar across methods. CBT may therefore offer a way to extend assistance to otherwise unserved communities while achieving comparable average welfare outcomes. This conclusion applies to the survey-based

PMT+ used in our experiment, not to all data-driven targeting technologies.

Committee design may also be adapted to address groups less well represented under CBT, including displaced households. Clearer eligibility guidance and safeguards could improve their inclusion, although broader coverage would reduce the associated cost savings. Testing such refinements would help establish whether selection can improve without compromising community relations or geographic reach.

Two further extensions would inform operational decisions. Repeated randomization across assistance cycles could assess whether committees continue to incorporate changing needs and whether favoritism emerges as households learn how selection works. Jointly randomizing targeting methods and coverage rates would distinguish the consequences of who receives assistance from those of how broadly it is distributed. Such evidence would help agencies choose both the selection method and the coverage level under a fixed budget.

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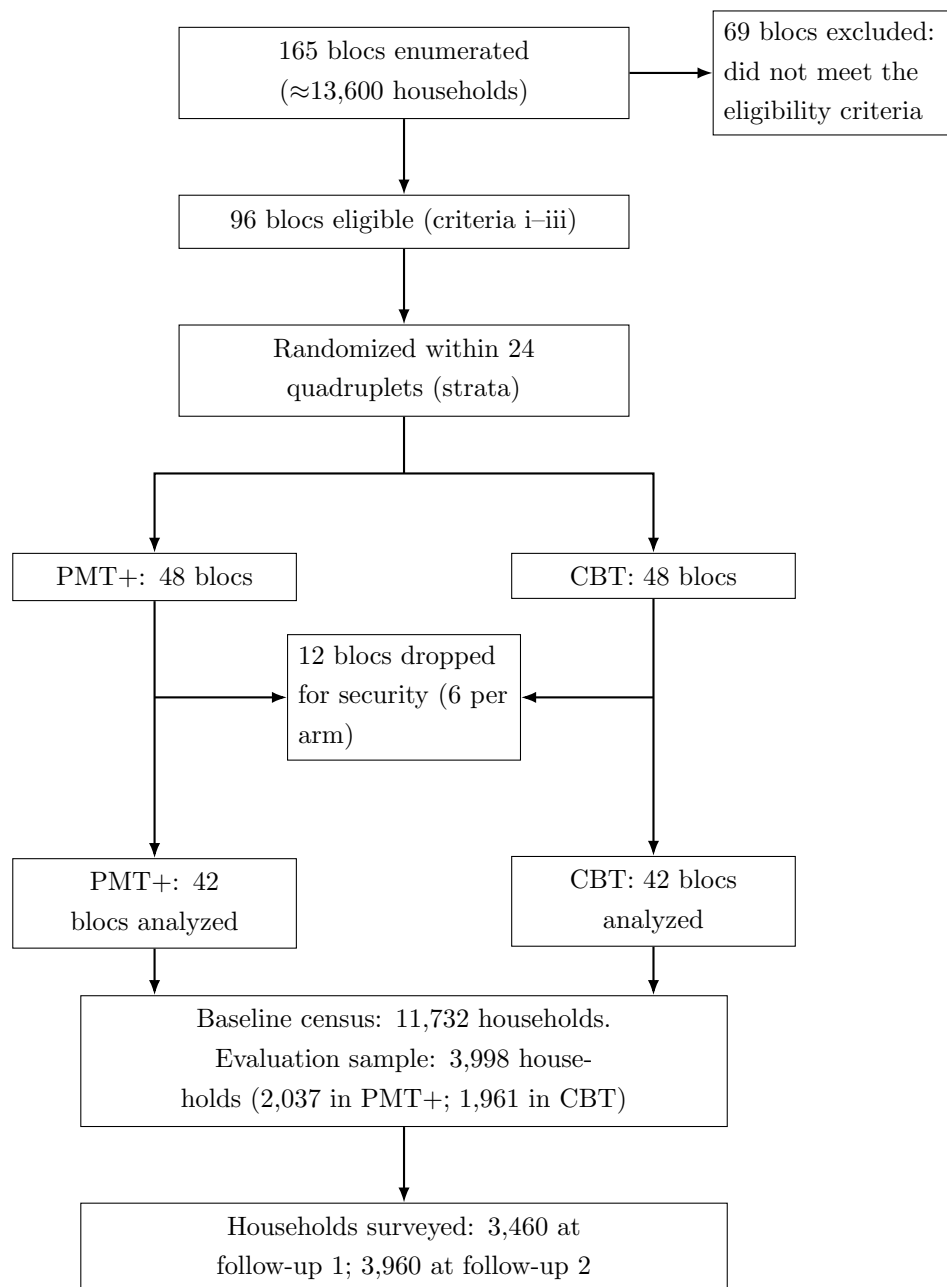
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A Additional Figures and Tables

Figure A1: Study design and participant flow



Notes: From 165 enumerated blocs, 96 met the eligibility criteria (i–iii) and were grouped into 24 quadruplets and randomly assigned within quadruplet to PMT+ or CBT; 12 blocs (six per arm) were dropped for security before the study began, leaving 84 analyzed blocs. The baseline census covered 11,732 households, from which an evaluation sample of 3,998 households was drawn and surveyed in two follow-up rounds. We stratified the 96 eligible blocs by geography and estimated population size to improve balance and statistical power and to limit potential power losses from bloc attrition. Within each of the three groupements, we first organized blocs by WFP’s operational axes—logistical corridors established for program implementation and transfer delivery—and then sorted them in descending order of household population, forming quadruplets of four adjacent, similarly sized blocs from which one was randomly assigned to each condition.

Table A1: Transfer value benchmarked against household expenditure

	Tanganyika
Households surveyed	1,641
Mean household size	6.03
<i>Per capita expenditure (CDF)</i>	
Mean, monthly	35,158
Median, monthly	30,130
Mean, six months	210,949
Median, six months	180,778
<i>Transfer as a share of expenditure (%)</i>	
Of the mean	60
Of the median	70

Notes: Computed from the WFP Emergency Food Security Assessment (EFSA) 2023, restricted to Tanganyika, the province in which the study takes place and the same year as the intervention. Monthly household expenditure sums food, short-term non-food, and long-term non-food expenditure, and is divided by household size to obtain the per capita figures. The transfer is 126,000 CDF per household member over six months, or 21,000 CDF per member per month; the shares in the final panel express that monthly transfer relative to monthly per capita expenditure.

Table A2: Target survey sample by targeting method and eligibility status

	Eligible	Ineligible	Total
PMT+	1,425	612	2,037
CBT	1,269	692	1,961
Total	2,694	1,304	3,998

Notes: Selected survey households across the 84 analyzed blocs. “Eligible” denotes households sampled as beneficiaries. The design targeted an equal split of beneficiaries and non-beneficiaries per bloc, over-sampling the scarcer non-beneficiary group.

Table A3: Baseline balance across targeting arms, 84 study blocs

	PMT+ (1)	CBT (2)	Difference (3)
<i>Panel A. Household demographics</i>			
Woman-headed household	0.13 (0.01)	0.13 (0.01)	−0.00 (0.01)
Household head age	39.29 (0.55)	40.00 (0.61)	−0.81* (0.48)
Head has no education	0.54 (0.03)	0.59 (0.03)	−0.05 (0.03)
Head is elderly (60+)	0.09 (0.01)	0.11 (0.01)	−0.02** (0.01)
Head has a disability	0.03 (0.01)	0.04 (0.00)	−0.01 (0.01)
Household size	6.17 (0.09)	6.21 (0.09)	−0.05 (0.10)
More than 3 children under 5	0.05 (0.01)	0.04 (0.01)	0.01 (0.01)
Household is Twa	0.21 (0.04)	0.17 (0.03)	0.05 (0.04)
Displaced household	0.04 (0.03)	0.08 (0.04)	−0.04 (0.03)
Lives from agriculture	0.57 (0.03)	0.55 (0.03)	0.02 (0.03)
Received WFP assistance in 2022	0.21 (0.04)	0.19 (0.04)	0.04 (0.04)
<i>Panel B. Welfare at baseline</i>			
Food Consumption Score	17.83 (0.52)	18.68 (0.62)	−0.72 (0.64)
Food insecure ($FCS \leq 28$)	0.88 (0.01)	0.86 (0.01)	0.02 (0.02)
Wealth index (PCA)	2.79 (0.10)	2.83 (0.10)	−0.06 (0.10)
Used crisis coping strategies	0.78 (0.02)	0.82 (0.02)	−0.04** (0.02)
Used emergency coping strategies	0.69 (0.02)	0.77 (0.02)	−0.07*** (0.02)
Households	2,037	1,961	
Blocs	42	42	
Joint orthogonality test (p -value)		0.005	

Notes: Columns (1) and (2) report unweighted arm means from the baseline census; standard errors clustered at the bloc level in parentheses. Column (3) is the PMT+−CBT difference from a regression on the CBT indicator with quadruplet fixed effects. Sample sizes vary slightly across rows due to item nonresponse (wealth index: $N = 1,784$ for PMT+ and $N = 1,736$ for CBT). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4: Attrition by study arm and follow-up round

	Follow-up 1		Follow-up 2	
	Before repl. (1)	After repl. (2)	Before repl. (3)	After repl. (4)
<i>Panel A. Households surveyed</i>				
Overall	3,200	3,460	3,637	3,960
PMT+	1,666	1,753	1,853	2,016
CBT	1,534	1,707	1,784	1,944
<i>Panel B. Attrition rate</i>				
Overall	0.200	0.135	0.090	0.010
PMT+	0.182	0.150	0.090	0.000
CBT	0.218	0.118	0.090	0.019
Differential (CBT – PMT+)	0.025	-0.001	0.003	0.010
<i>p</i> -value	0.466	0.980	0.876	0.328

Notes: Attrition rates relative to the target sample, before and after replacement households, at each follow-up. The differential is the CBT–PMT+ gap; the *p*-value is from a bloc-level regression of the attrition indicator on the CBT indicator with quadruplet fixed effects, clustered at the bloc level.

Table A5: Transfer take-up and adherence (SCOPE administrative records)

	Eligible			Non-eligible
	Overall	PMT+	CBT	
<i>Panel A. Adherence</i>				
Received at least one transfer	0.93	0.94	0.92	0.00
Received all four transfers	0.82	0.84	0.81	0.00
<i>Panel B. Number of transfers received</i>				
Four transfers	0.82	0.84	0.81	0.00
Three transfers	0.01	0.01	0.01	0.00
Two transfers	0.09	0.09	0.10	0.00
One transfer	0.00	0.00	0.00	0.00
No transfers	0.07	0.06	0.08	1.00

Non-eligible households receiving any transfer: 1 of 1,294.

Notes: Shares computed from WFP SCOPE administrative records over the analysis sample with known baseline eligibility. Panel A reports the share of eligible households that received at least one and all four transfers; Panel B gives the distribution of eligible households by the total number of transfers received. Columns report the eligible sample overall and separately by targeting arm. Households with no SCOPE record are treated as non-receipt.

Table A6: Characteristics of selected households: CBT versus PMT+ at equal coverage

Characteristic	CBT	PMT+	Difference
Household head is illiterate	57.5	62.6	-5.2***
Household living from agriculture (farm-headed)	47.6	42.9	+4.6***
Agricultural household with low production	28.3	24.9	+3.4***
Household has a pregnant or lactating woman	32.8	29.8	+3.0*
Household lost crops to insects	18.4	16.0	+2.4***
Displaced household	38.8	40.9	-2.1**
Twa (indigenous) household	13.6	11.6	+2.0*
Female-headed household	19.4	20.6	-1.2*
Host family	6.5	5.3	+1.2*
Household head is chronically ill	3.8	4.0	-0.2**

Notes: Difference in the share of households with each characteristic that are selected under CBT versus under the PMT+ rule simulated at the same coverage within CBT blocs. Positive values indicate the characteristic is more common among CBT-selected households. The ten characteristics with the largest statistically significant differences are shown. Computed on the full bloc census, so no sampling weight applies. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A7: Relative representation of vulnerable subgroups by targeting method

Subgroup	RR under PMT+	RR under CBT	Difference
Agriculture-dependent (farm-headed) household	0.95	1.02	+0.07*
Agricultural household with low production	0.90	0.99	+0.09
Household has a pregnant or lactating woman	0.95	1.20	+0.25*
Household has a member with a disability	0.93	1.13	+0.20
Illiterate household head	1.18	1.02	-0.16***
Widow-headed household	1.14	1.19	+0.05
Displaced household	0.44	0.26	-0.18
Household has an orphaned child	1.33	1.38	+0.05
Elderly-headed household	1.31	1.48	+0.18
Twa (indigenous) household	0.81	0.92	+0.11

Notes: The relative-representation index RR_g is the subgroup's share among beneficiaries divided by its share among the food insecure ($FCS \leq 28$). $RR_g = 1$ denotes representation proportional to the group's presence among the needy, below one under-representation. Columns report RR_g under PMT+ and CBT and their difference. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A8: Do committees select on shocks the model misses?

	CBT selection (eligibility)
Crop loss – insects	0.025 (0.027)
Crop loss – natural	-0.016 (0.035)
HoH chronically ill	0.112*** (0.036)
HoH disabled	0.257*** (0.070)
Displaced household	0.256*** (0.074)
PMT signal (predicted FCS)	Yes
Quadruplet FE	Yes

Notes: Linear probability regression, within CBT blocs, of an indicator for CBT selection on recent-shock and vulnerability indicators, conditioning on the PMT predicted-FCS score and quadruplet fixed effects; standard errors clustered at the bloc level. Positive coefficients indicate committees select on a dimension over and above what the statistical score already captures. Computed on the full bloc census, so no sampling weight applies. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A9: Targeting precision under the wealth-index benchmark

	Coverage free		Equal coverage		Coverage controlled	
	PMT+	CBT	PMT+	CBT	PMT+	CBT
Exclusion error	0.182	0.273** (0.040)	0.274	0.273 (0.010)	0.235	0.219 (0.016)
Inclusion error	0.259	0.292 (0.033)	0.279	0.292 (0.011)	0.279	0.271 (0.030)
CGH index	0.866	0.934 (0.044)	0.943	0.934 (0.016)	0.871	0.929 (0.038)
CGH (bottom 10%)	0.989	1.103 (0.129)	0.911	1.103 (0.143)	1.037	1.055 (0.097)
CGH (bottom 20%)	0.990	1.077 (0.076)	1.011	1.077 (0.052)	1.019	1.048 (0.061)
CGH (bottom 40%)	0.994	1.070 (0.045)	1.055	1.070 (0.027)	1.007	1.057 (0.033)
Targeting Differential	0.116	0.052 (0.049)	0.105	0.052 (0.045)	0.107	0.061 (0.043)
TD (bottom 10%)	0.039	0.118 (0.054)	0.030	0.118 (0.058)	0.048	0.109 (0.061)
TD (bottom 20%)	0.047	0.095 (0.041)	0.051	0.095 (0.037)	0.053	0.088 (0.043)
TD (bottom 40%)	0.058	0.087 (0.036)	0.077	0.087 (0.031)	0.064	0.081 (0.032)
Number of blocs	84		42		84	

Notes: The table repeats Table 1 against an asset-based wealth benchmark: households are ranked from poorest and the bottom 74.6%, the average coverage across the two arms, are classified as in need; Appendix B defines each measure and the index. Column blocks are as in Table 1. Standard errors clustered at the bloc level in parentheses; all regressions include quadruplet fixed effects. Stars refer to the CBT–PMT+ difference. Restricting the index to the seven asset-ownership indicators, dropping housing type and the sanitation, water, and cooking-fuel measures, leaves the aggregate rows substantively unchanged. The bottom- X -percent rows are sensitive both to index composition and to how ties are broken, and we draw no conclusion from them; see Section 5. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A10: Heterogeneity in targeting precision (CBT $\times$ bloc moderators)

	Moderator: CV of FCS		Moderator: food-insecure share	
	No cov. control (1)	With cov. control (2)	No cov. control (3)	With cov. control (4)
<i>Panel A. Exclusion error</i>				
CBT	0.310 (0.223)	0.087 (0.065)	-0.507 (0.336)	-0.189 (0.238)
CBT $\times$ moderator	-0.456 (0.471)	-0.170 (0.143)	0.703* (0.390)	0.231 (0.273)
<i>Panel B. Inclusion error</i>				
CBT	-0.117 (0.124)	-0.114 (0.134)	-0.149 (0.242)	-0.182 (0.240)
CBT $\times$ moderator	0.290 (0.259)	0.287 (0.273)	0.174 (0.273)	0.224 (0.275)
Quadruplet FE	Yes	Yes	Yes	Yes
Coverage control	No	Yes	No	Yes
Number of blocs	84	84	84	84

Notes: Bloc-level regressions of the exclusion (Panel A) and inclusion (Panel B) error rate on the CBT indicator, a bloc moderator, and their interaction, with quadruplet fixed effects and standard errors clustered at the bloc level; unweighted. The moderator is the within-bloc coefficient of variation of the FCS (columns 1–2) or the share of food-insecure households (columns 3–4); the even columns add the bloc coverage rate as a control. Computed on the full bloc census, so no sampling weight applies. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A11: Effect of CBT on household welfare, census-calibrated population weights

	Follow-up 1		Follow-up 2	
	PMT+ mean (1)	CBT effect (2)	PMT+ mean (3)	CBT effect (4)
Food Consumption Score	19.910	-2.013* (1.206)	14.327	0.528 (0.468)
Reduced Coping Strategies Index (rCSI)	22.222	-0.587 (0.855)	25.670	0.303 (0.731)
Used stress coping strategy	0.965	-0.003 (0.011)	0.934	0.024*** (0.008)
Used crisis coping strategy	0.743	-0.001 (0.024)	0.778	0.003 (0.020)
Used emergency coping strategy	0.605	0.040 (0.045)	0.735	-0.034* (0.019)
Life satisfaction (Anderson index)	-0.131	0.153 (0.094)	0.187	-0.038 (0.071)
Women's agency (Anderson index)	-0.073	0.061 (0.055)	0.032	0.081** (0.037)

Notes: Intention-to-treat effect of CBT on each household-welfare outcome, reweighting the sample to the bloc census with the census-calibrated population weight (the within-quadruplet design weight raked to census margins, winsorized at the 99th percentile); household-level regressions with quadruplet fixed effects, double-LASSO-selected baseline controls, and bloc-clustered standard errors. The conclusions are unchanged under population weighting. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A12: Equivalence: point estimates, confidence intervals, and minimum detectable effects

	CBT effect $\hat{\tau}$ (1)	95% CI (2)	MDE (units) (3)	MDE / control SD (4)
Food Consumption Score (FU1)	0.222	[-1.004, 1.447]	1.751	0.180
Food Consumption Score (FU2)	0.271	[-0.396, 0.938]	0.953	0.087
Reduced Coping Strategies Index (FU1)	0.434	[-0.649, 1.517]	1.547	0.144
Reduced Coping Strategies Index (FU2)	0.092	[-0.995, 1.179]	1.553	0.158
Life satisfaction (Anderson index, FU2)	-0.023	[-0.101, 0.055]	0.111	0.111
Non-beneficiary FCS spillover (FU1)	1.615	[-0.081, 3.311]	2.423	0.273
Non-beneficiary FCS spillover (FU2)	0.293	[-0.526, 1.111]	1.170	0.112

Notes: For each welfare outcome, the CBT ITT estimate, its 95% confidence interval, the minimum detectable effect ($MDE = 2.8 \times SE$), and the MDE as a fraction of the PMT+ control-group standard deviation. Small MDE/SD ratios indicate the nulls are precisely estimated rather than underpowered.

Table A13: Heterogeneity in welfare effects by bloc characteristics

	Bloc size (>60 households)		Coverage rate (>76%)		Food-insecure share (>88%)	
	Unwtd. (1)	Wtd. (2)	Unwtd. (3)	Wtd. (4)	Unwtd. (5)	Wtd. (6)
<i>Panel A. Follow-up 1</i>						
Food Consumption Score	-2.465* (1.348)	-3.525** (1.369)	0.707 (1.469)	-2.755** (1.361)	1.098 (1.244)	-1.133 (2.358)
Reduced Coping Strategies Index (rCSI)	2.135* (1.166)	-0.447 (1.284)	1.548 (1.417)	-1.190 (1.935)	0.803 (1.152)	-0.054 (1.667)
Used stress coping strategy	0.036* (0.019)	0.028 (0.019)	0.051* (0.026)	0.082*** (0.026)	0.021 (0.019)	-0.009 (0.028)
Used crisis coping strategy	0.090** (0.043)	0.093** (0.044)	0.019 (0.057)	0.040 (0.046)	0.037 (0.041)	0.044 (0.062)
Used emergency coping strategy	0.100 (0.076)	0.165** (0.079)	0.152* (0.085)	0.233*** (0.065)	0.040 (0.076)	0.009 (0.068)
Life satisfaction (Anderson index)	-0.112 (0.124)	0.119 (0.147)	-0.130 (0.147)	0.157 (0.169)	-0.102 (0.111)	-0.176 (0.138)
Women's agency (Anderson index)	0.060 (0.074)	0.050 (0.089)	0.110 (0.097)	0.293*** (0.107)	-0.041 (0.081)	0.011 (0.090)
<i>Panel B. Follow-up 2</i>						
Food Consumption Score	-1.457** (0.678)	-0.496 (0.791)	-0.625 (0.804)	-1.202 (0.931)	0.193 (0.683)	-0.794 (1.069)
Reduced Coping Strategies Index (rCSI)	0.463 (1.164)	0.259 (1.310)	3.485*** (1.209)	3.583*** (1.202)	1.107 (1.273)	2.785*** (1.045)
Used stress coping strategy	0.029** (0.012)	0.051*** (0.013)	0.000 (0.015)	0.012 (0.015)	-0.019 (0.015)	-0.002 (0.015)
Used crisis coping strategy	0.006 (0.029)	-0.015 (0.032)	0.037 (0.034)	0.012 (0.040)	0.077*** (0.028)	0.092*** (0.030)
Used emergency coping strategy	-0.020 (0.044)	-0.033 (0.035)	-0.044 (0.041)	-0.064 (0.049)	-0.008 (0.041)	0.012 (0.048)
Life satisfaction (Anderson index)	-0.053 (0.084)	-0.042 (0.111)	0.057 (0.097)	0.228* (0.138)	0.255*** (0.099)	0.467*** (0.113)
Women's agency (Anderson index)	0.088 (0.070)	0.132* (0.070)	-0.011 (0.081)	0.046 (0.088)	-0.079 (0.068)	-0.114 (0.085)
<i>Panel C. Perceived safety (follow-up 2)</i>						
Felt unsafe at home	0.129** (0.058)	0.197*** (0.075)	0.015 (0.060)	0.183** (0.086)	0.020 (0.059)	-0.068 (0.099)
Felt unsafe in community	0.066 (0.054)	0.121* (0.068)	0.008 (0.058)	0.090 (0.081)	0.034 (0.053)	-0.062 (0.081)
Quadruplet FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of blocs				84		

Notes: Each cell is the interaction coefficient $CBT \times H$ from a household-level regression of the outcome on the CBT indicator, an above-median bloc moderator H , and their interaction, with quadruplet fixed effects, double-LASSO-selected baseline controls, and standard errors clustered at the bloc level (in parentheses). The three moderators are bloc size, realized coverage, and the baseline food-insecure share, each split at its median; ties at the median make the splits uneven (35/49, 40/44 and 41/43 blocs respectively). Unweighted columns are the paper's primary specification; weighted columns reweight to the bloc census with the calibrated population weight. Panels A and B report the seven welfare outcomes of Table 2 at each follow-up; Panel C reports the two perceived-safety items, measured only at the second follow-up. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A14: Difference-in-differences: effect of the cash transfer on FCS

	Food Consumption Score
Follow-up 1 (non-beneficiary trend)	-1.97 (0.38)
Follow-up 2 (non-beneficiary trend)	-4.80 (0.39)
Follow-up 1 $\times$ Beneficiary (transfer effect)	3.02*** (0.47)
Follow-up 2 $\times$ Beneficiary (transfer effect)	1.92*** (0.47)
Household fixed effects	Yes
Observations	10951

Notes: Household-level regression of the FCS on survey-round indicators, a baseline-beneficiary indicator, and their interactions, with household fixed effects and standard errors clustered at the household level; unweighted. The follow-up $\times$ beneficiary interactions are the difference-in-differences estimates of the transfer's effect. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A15: Effect of CBT on household welfare, controlling for coverage

	Follow-up 1		Follow-up 2	
	PMT+ mean (1)	CBT effect (2)	PMT+ mean (3)	CBT effect (4)
Food Consumption Score	18.118	-0.197 (0.712)	14.561	-0.059 (0.320)
Reduced Coping Strategies Index (rCSI)	21.182	0.459 (0.636)	25.198	0.634 (0.555)
Used stress coping strategy	0.966	-0.016 (0.011)	0.952	-0.003 (0.006)
Used crisis coping strategy	0.754	-0.035 (0.023)	0.774	0.016 (0.014)
Used emergency coping strategy	0.618	-0.034 (0.039)	0.711	-0.040** (0.020)
Life satisfaction (Anderson index)	-0.031	0.030 (0.060)	0.004	-0.040 (0.045)
Women's agency (Anderson index)	-0.029	0.075* (0.041)	-0.010	0.017 (0.031)

Notes: Replicates Table 2 adding the bloc coverage rate (the share of bloc households selected for assistance) as a control. Because CBT covered fewer households than PMT+, the CBT coefficient here compares the two methods at the same coverage rate, separating the selection method from the size of the caseload. Columns (1) and (3) report the PMT+ mean at the first and second follow-up; columns (2) and (4) report the CBT effect with its standard error below. Unweighted household-level regressions with quadruplet fixed effects, double-LASSO-selected baseline controls, and standard errors clustered at the bloc level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A16: Welfare of the baseline food-insecure, by targeting method

	PMT+ baseline-FI mean (1)	CBT effect (Follow-up 1) (2)	CBT effect (Follow-up 2) (3)
Food Consumption Score	14.488	-0.218 (0.644)	0.224 (0.340)
Reduced Coping Strategies Index (rCSI)	25.161	0.450 (0.576)	0.367 (0.557)
Used stress coping strategy	0.950	-0.010 (0.009)	0.001 (0.007)
Used crisis coping strategy	0.769	-0.030 (0.021)	0.020 (0.016)
Used emergency coping strategy	0.709	-0.044 (0.037)	-0.019 (0.018)
Life satisfaction (Anderson index)	-0.022	0.051 (0.057)	-0.026 (0.040)
Women's agency (Anderson index)	-0.016	0.049 (0.040)	0.026 (0.035)

Notes: ITT effect of CBT on the subgroup that was food insecure at baseline (census FCS ≤ 28), a group fixed before treatment, avoiding the post-treatment conditioning of the non-beneficiary spillover table. Unweighted, quadruplet fixed effects, double-LASSO controls, bloc-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A17: Effect of CBT on social cohesion and perceived safety, census-calibrated population weights

	Follow-up 1		Follow-up 2	
	PMT+ mean (1)	CBT effect (2)	PMT+ mean (3)	CBT effect (4)
<i>Trust and social cohesion</i>				
Social cohesion index	–	–	0.106	-0.101 (0.080)
<i>Perceived safety (share who ever felt unsafe)</i>				
Felt unsafe at home	–	–	0.468	0.148*** (0.050)
Felt unsafe in community	–	–	0.549	0.094** (0.038)

Notes: Intention-to-treat effect of CBT on social cohesion and perceived-safety outcomes, reweighted to the bloc census with the census-calibrated population weight (the within-quadruplet design weight raked to census margins, winsorized at the 99th percentile); quadruplet fixed effects, double-LASSO controls, bloc-clustered standard errors. Social cohesion remains insignificant. The perceived-insecurity effects strengthen under weighting—the CBT effect on ever feeling unsafe at home rises to 0.148 ($p < 0.01$) and in the community to 0.094 ($p < 0.05$). We read the unweighted estimates as primary. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A18: Targeting-related cases in the Complaints and Feedback Mechanism

	PMT+ mean (1)	CBT effect	
		No coverage (2)	With coverage (3)
Received at least one targeting-related case	0.238	-0.019 (0.074)	-0.017 (0.088)
Number of targeting-related cases	1.786	-0.044 (1.162)	0.074 (1.094)
Cases per 100 census households	3.352	-0.224 (2.090)	-0.046 (1.979)
Cases per 100 non-selected households	4.355	5.684 (5.808)	7.062 (8.107)
Quadruplet FE		Yes	Yes
Coverage control		No	Yes
Number of blocs		84	

Notes: Bloc-level effects of assignment to CBT on cases logged in WFP's Complaints and Feedback Mechanism during the distribution cycle. Of 651 cases recorded across the 84 study blocs, 148 are targeting-related: 43 allege exclusion or report a larger household, 7 allege inclusion or report a smaller household, and 98 are requests for assistance from households that were not selected. The remaining 503 concern the mechanics of collection—an absent or replaced household member, or a problem with a token or identity document—and are excluded here. Column (1) is the PMT+ mean; columns (2) and (3) report the CBT effect without and with the bloc coverage rate as a control, each from a bloc-level regression with quadruplet fixed effects and standard errors clustered at the bloc level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A19: Favoritism: perceptions, committee connection, and selection

<i>Panel A: effect of CBT (both arms)</i>				
	PMT+ mean (1)	Unadjusted (2)	Coverage-adjusted (3)	Weighted (4)
Some households have better connections	0.025	0.014* (0.008)	0.013 (0.009)	0.008 (0.009)
Household was part of the committee	0.199	-0.017 (0.020)	-0.010 (0.022)	-0.060 (0.040)
Knows a committee member	0.205	0.026 (0.019)	0.020 (0.021)	0.061* (0.033)
<i>Panel B: does connection predict selection? (CBT blocs)</i>				
		Unconditional	Conditional on need	
Household member on the committee		-0.017 (0.034)	-0.017 (0.034)	
Knows a committee member		0.007 (0.040)	0.007 (0.040)	

Notes: Panel A reports the effect of assignment to CBT on three favoritism-related items, all measured in both arms at the second follow-up. They are shown as separate rows rather than combined into an index because their pairwise correlations range from 0.002 to 0.081, so they do not represent a common construct. Column (1) is the PMT+ mean; columns (2)–(4) give the CBT effect unadjusted, adding the bloc coverage rate, and reweighted to the bloc census, each from equation (6) with quadruplet fixed effects, double-LASSO controls, and bloc-clustered standard errors. Panel B reports the direct test of elite capture, estimated within CBT blocs only: an indicator for selection regressed on each connection measure, with quadruplet fixed effects and bloc-clustered standard errors, without and with a control for baseline need (census FCS ≤ 28). Because selection under CBT is defined only in CBT blocs, Panel B has no PMT+ counterfactual and is not an intention-to-treat estimate. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A20: When does not targeting dominate? Prevalence thresholds

λ	Rule 1: per bloc		Rule 2: regional budget	
	PMT+	CBT	PMT+	CBT
<i>Panel A. Prevalence threshold π^* (blanket dominates for $\pi_b > \pi^*$)</i>				
0.5	0.686	0.566	0.863	0.824
0.7	0.412	0.159	0.859	0.831
0.8	0.071	-0.350	0.855	0.841
0.9	-0.954	-1.878	0.842	0.870
<i>Panel B. Share of the 84 blocs where blanket dominates (%)</i>				
0.5	95	99	54	73
0.7	100	100	54	68
0.8	100	100	55	63
0.9	100	100	63	51

Notes: π^* is the bloc prevalence above which universal assistance yields a lower calculated loss than targeting method m , from equation (7). Rule 1 compares within a single funded bloc; Rule 2 adds the cost premium of universal coverage ($\rho = 1.23$ for PMT+, 1.42 for CBT). Panel B gives the share of the 84 blocs whose observed prevalence exceeds π^* ; a negative π^* means universal assistance dominates at any prevalence.

Table A21: Value of information: targeting versus random rationing at equal coverage

λ	PMT+	CBT
<i>Panel A. Mean value of information (planner-loss units)</i>		
0.5	0.004	0.011
0.7	0.001	0.007
0.8	-0.001	0.005
0.9	-0.003	0.003
<i>Panel B. Share of blocs where targeting beats random (%)</i>		
0.5	40	38
0.7	40	38
0.8	38	36
0.9	36	33

Notes: Each method is compared with random rationing of the same number of transfers in the same bloc, holding the caseload fixed so only the ranking differs. Panel A reports realized errors against those of a random draw and the resulting value of information—the reduction in the loss of equation (7), net of screening cost—at each λ . Panel B gives the share of blocs in which the method improves on random rationing.

The remaining household-level intention-to-treat exhibits are repeated under the census-calibrated population weight: non-beneficiary spillovers (Table A24), satisfaction and perceptions of the targeting process (Table A25), and social cohesion and perceived safety (Table A17). The paper’s central conclusions are robust: welfare parity and the absence of harm to non-beneficiaries both hold. Several second-order, component-level results move under weighting—the marginal “less fair” rating loses significance while the “slower process” and lower-collective-action findings persist—which is one reason the main text reports these outcomes as summary indices rather than item by item (Table 4). Because assignment is randomized within quadruplets, we treat the unweighted estimates as primary (Section 4) and read these as a reweighting robustness check.

Table A22: Effect of CBT on satisfaction and perceptions of the targeting process

	Follow-up 2	
	PMT+ mean (1)	CBT effect (2)
<i>Perceptions of the targeting process</i>		
Satisfied with targeting process	0.607	0.005 (0.025)
Views process as fair	0.411	-0.040* (0.022)
Views process as transparent	0.749	0.003 (0.020)
Views process as corrupt	0.005	0.000 (0.003)
Views process as slow	0.199	0.059*** (0.015)

Notes: Components of the satisfaction index in Table 4, each measured at the second follow-up. Column (1) is the PMT+ mean; column (2) is the intention-to-treat effect of CBT from equation (6), unweighted, with quadruplet fixed effects, double-LASSO controls and standard errors clustered at the bloc level. Table A25 reports the reweighted twin. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A23: Effect of CBT on social cohesion and perceived safety

	Follow-up 1		Follow-up 2	
	PMT+ mean (1)	CBT effect (2)	PMT+ mean (3)	CBT effect (4)
<i>Trust and social cohesion</i>				
Social cohesion index	–	–	-0.008	0.020 (0.041)
<i>Perceived safety (share who ever felt unsafe)</i>				
Felt unsafe at home	–	–	0.546	0.040 (0.027)
Felt unsafe in community	–	–	0.573	0.041* (0.025)

Notes: The safety items underlying the tension index in Table 4, reported separately rather than pooled, alongside the social cohesion index. The cohesion row reproduces the combined index of Table 4 (which folds in the eight trust items) for reference next to its components. Cohesion and the safety items were collected only at the second follow-up. Columns (1) and (3) are PMT+ means and columns (2) and (4) the intention-to-treat effect of CBT from equation (6), unweighted, with quadruplet fixed effects, double-LASSO controls and standard errors clustered at the bloc level. Table A17 reports the reweighted twin. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A24: Effect of CBT on non-beneficiary welfare, census-calibrated population weights

	PMT+ non-benef. mean (1)	CBT effect (Follow-up 1) (2)	CBT effect (Follow-up 2) (3)
Food Consumption Score	14.180	1.483 (1.059)	0.300 (0.679)
Reduced Coping Strategies Index (rCSI)	24.766	-0.115 (1.106)	1.026 (0.667)
Used stress coping strategy	0.965	-0.025 (0.015)	-0.010 (0.013)
Used crisis coping strategy	0.757	-0.047 (0.045)	0.010 (0.037)
Used emergency coping strategy	0.726	0.038 (0.042)	-0.075* (0.041)
Life satisfaction (Anderson index)	0.075	-0.044 (0.114)	-0.051 (0.053)
Women's agency (Anderson index)	-0.068	0.186** (0.086)	-0.016 (0.055)

Notes: Intention-to-treat effect of CBT on non-beneficiary households, reweighted with the census-calibrated population weight (the within-quadruplet design weight raked to census margins, winsorized at the 99th percentile); quadruplet fixed effects, double-LASSO controls, bloc-clustered standard errors. The central finding—no systematic evidence that CBT's lower coverage harms the households it leaves out—is unchanged: the two nominally significant cells in the unweighted specification (the first-follow-up FCS and second-follow-up rCSI) lose significance under weighting, and a few coping-strategy cells become marginally significant with no consistent direction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A25: Effect of CBT on satisfaction and perceptions of the targeting process, census-calibrated population weights

	Follow-up 2	
	PMT+ mean (1)	CBT effect (2)
<i>Perceptions of the targeting process</i>		
Satisfied with targeting process	0.644	0.017 (0.038)
Views process as fair	0.394	0.023 (0.061)
Views process as transparent	0.733	0.006 (0.013)
Views process as corrupt	0.012	-0.009** (0.004)
Views process as slow	0.173	0.091*** (0.029)

Notes: Intention-to-treat effect of CBT on satisfaction and perceptions of the targeting process, reweighted to the bloc census with the census-calibrated population weight (the within-quadruplet design weight raked to census margins, winsorized at the 99th percentile); quadruplet fixed effects, double-LASSO controls, bloc-clustered standard errors. Satisfaction remains high and flat and the “slower process” perception persists and grows (0.091, $p < 0.01$); the lower “fair” rating from the unweighted specification turns small and insignificant. We read the unweighted estimates, consistent with within-quadruplet randomization, as primary. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B Outcome and Performance Measures

This appendix defines every measure the paper reports, in the order the results present them. Table B1 lists the component survey items of the five composite indices.

Targeting performance. The **coverage rate** is the share of households in a bloc selected for assistance, averaged across blocs within each arm. The **exclusion error rate** is, among food-insecure households (census FCS at or below 28), the share left unselected—the number of unselected food-insecure households divided by the number of food-insecure households. The **inclusion error rate** is, among selected households, the share that are not food insecure (census FCS above 28)—the number of selected non-food-insecure households divided by the number of selected households (Aiken et al., 2022). The **Coady–Grosh–Hoddinott (CGH) index** is a distribution-sensitive measure of progressivity: the share of transfers accruing to food-insecure households relative to their population share, where one denotes a neutral allocation and values above one indicate progressive targeting (Coady et al., 2004). The **Targeting Differential (TD)** is the difference in program participation rates between food-insecure and non-food-insecure households, ranging from -1 to 1 and equal to zero under proportional targeting (Galasso and Ravallion, 2005). CGH and TD are also evaluated at the bottom 10%, 20%, and 40% of the benchmark distribution; because both benchmarks are heavily tied at the bottom, these tail sets are not unique and we do not treat the tail rows as identified. The **relative-representation index (RR)** of a subgroup is its share of all beneficiaries divided by its share of all food-insecure households: one denotes representation exactly in proportion to presence among the food insecure, below one underrepresentation, and above one overrepresentation. The **wealth index**, an alternative benchmark for need, is constructed from the same census as the first principal component (Filmer and Pritchett, 2001) of 11 variables capturing asset ownership, housing type, and access to sanitation, water, and cooking fuel.

Household welfare. Table 2 reports seven outcomes. The **Food Consumption Score (FCS)** aggregates dietary diversity and consumption frequency over the previous seven days into a score from 0 to 112: the number of days in the week each of eight food groups was consumed—staples, pulses, vegetables, fruit, meat and fish, dairy, sugar, and oil—weighted by the group’s nutritional value, with higher values indicating better food consumption (WFP, 2008). The **reduced Coping Strategies Index (rCSI)** captures the frequency and severity of consumption-related coping as the sum over the past seven days of the days each of five behaviors was used, weighted by a severity factor: relying on less preferred or less expensive food (1), borrowing food or relying on help from friends or relatives (2),

reducing the number of meals per day (1), reducing portion size (1), and restricting adult consumption so that young children can eat (3). The index ranges from 0 to 56, with higher values worse (Maxwell and Caldwell, 2008). **Livelihood coping** is measured by three binary indicators, each equal to one if the household used at least one strategy in the corresponding severity category (WFP, 2021): *stress* (sold household assets, spent savings, borrowed money or food from a lender or bank, or sent household members to eat elsewhere), *crisis* (sold productive assets or means of transport, reduced essential non-food expenditure on health or education, or took children out of school), and *emergency* (sold the house or land, or begged for money). The module also elicits sales of livestock, which is not assigned to a severity category. The recall period is the past seven days at the first follow-up and the past 30 days at the second, so the two columns are not comparable in levels; the treatment contrast within each column is unaffected. The **life satisfaction index** is an inverse-covariance-weighted index of three Cantril-ladder items (each 1–10) on perceived quality of life, respect within the community, and how widely the respondent’s opinion is valued, with higher values indicating greater satisfaction (Cantril, 1965); its components are listed in Table B1. The **women’s agency index**, following Jayachandran et al. (2023), is an inverse-covariance-weighted index of eight items on the respondent’s decision-making power and freedom of movement, listed in Table B1.

The targeting process. The **satisfaction index** is an inverse-covariance-weighted index of four items measured at the second follow-up: whether the respondent is satisfied with the targeting process, and whether the respondent describes the selection process as fair, as transparent, and as *not* slow (the slow indicator is reversed so that all components run in the same direction). The satisfaction component is itself a binary indicator equal to one if the respondent reports being satisfied or very satisfied on at least one of two Likert-scale questions, one on the overall beneficiary-selection process and one on the formation of the local targeting mechanism. Respondents also select up to three adjectives from a predefined list to characterize the selection process; Table A22 reports overall satisfaction alongside the fair, transparent, corrupt, and slow indicators individually. *Corrupt* is deliberately excluded from the index: it is near-constant (a mean of 0.006, about 24 positive responses), so it carries no information for an inverse-covariance weighting and destabilizes the weights. It is reported on its own instead.

Community relations. The **social cohesion index**, adapted from Bossuoy et al. (2022), is an inverse-covariance-weighted index of sixteen items spanning four dimensions of community life: closeness and mutual support, community tension, collective action, and trust

in eight named groups. Trust enters at the item level rather than as an index of an index, so the eight trust items are weighted alongside the other eight components; Table B1 lists them all. The **tension index** is an inverse-covariance-weighted index of three items oriented so that higher values indicate more tension: having felt unsafe at home, having felt unsafe in the bloc, and having experienced at least one conflict in the past 30 days. Two further candidates are excluded and reported separately rather than folded in—whether the respondent reports hatred between groups (25% item nonresponse and weakly correlated with the rest) and whether the targeting process itself caused tension (only 2.3% positive and near-orthogonal to the others). Community tension over the past six months enters the cohesion index rather than the tension index, following Bossuoy et al. (2022), and is therefore not double-counted. **Perceived safety** is reported separately in Table A23 through the index’s two safety components: binary indicators equal to one if the respondent reports having felt unsafe at least occasionally in the past six months, at home and in the community respectively. For **favoritism** we use three measures: whether the respondent believes some households have better connections, whether a household member served on the targeting committee, and whether the household knows a committee member.

Table B1: Detailed description of the composite indices

Index	Component survey items	Response scale
Life satisfaction index	Self-anchoring (Cantril) ladder: (i) current quality of life; (ii) respect received within the community; (iii) how widely the respondent’s opinion is valued.	Ladder 1–10; higher values indicate greater satisfaction
Women’s agency index	(i) opinion heard when an expensive item is purchased; (ii) no permission needed to buy clothing; (iii) allowed to go to the market without asking; (iv) allowed to visit or talk to other women; (v) allowed to travel by public transport; (vi) allowed to go alone to meet friends; (vii) decides alone on emergency medicine for a child; (viii) allowed to go alone to a relative’s house in the village.	Likert 1–6, except (ii) binary; higher values indicate greater agency
Satisfaction index	(i) satisfied with the beneficiary-selection process or with how the local targeting mechanism was formed; (ii) describes the process as fair; (iii) describes the process as transparent; (iv) does <i>not</i> describe the process as slow.	Binary; higher values indicate greater satisfaction
Social cohesion index	<i>Closeness and support</i> : (i) can count on help from the bloc in times of difficulty; (ii) how many of ten people in the bloc are honest and trustworthy; (iii) closeness to the bloc community. <i>Tension</i> : (iv) tension among bloc members in the past six months. <i>Collective action</i> : (v) number of groups or associations belonged to; (vi) positions of responsibility held; (vii) money donated to community causes; (viii) days volunteered. <i>Trust</i> : trust in (ix) extended family; (x) immediate neighbors; (xi) people in the bloc; (xii) the bloc chief; (xiii) the government; (xiv) foreigners; (xv) NGOs; (xvi) other ethnic groups.	Counts and Likert scales (trust items 1–5)
Tension index	(i) felt unsafe at home at least occasionally in the past six months; (ii) felt unsafe in the bloc at least occasionally in the past six months; (iii) experienced at least one conflict in the past 30 days.	Binary; higher values indicate more tension

Notes: Each index is an inverse-covariance-weighted average of its standardized components, following [Anderson \(2008\)](#): every component is standardized to mean zero and unit standard deviation, aggregated with weights equal to the inverse of the components’ covariance matrix, and the composite re-standardized. Components are oriented before aggregation so that higher values denote greater life satisfaction, greater women’s agency, greater satisfaction, greater social cohesion, and more tension. The life satisfaction and women’s agency indices are welfare outcomes reported in Table 2. Three candidate components are excluded and reported separately—“describes the process as corrupt” from the satisfaction index, and reported hatred between groups and targeting-induced tension from the tension index—because each is near-constant or weakly correlated with the remaining items; Appendix B gives the details.

C Comparing Error Rates with the Literature

Published targeting errors use different definitions, so the levels in Table C1 require careful interpretation. Four measures appear in the literature: the share of selected households not in need (one minus precision, our inclusion error), the share of households in need not selected (one minus recall, our exclusion error), the share of households not in need who are selected (the false-positive rate), and the share of all households misclassified. Inclusion and exclusion errors coincide when program coverage equals the share of the population deemed in need. Several studies impose this equality by construction. It does not hold in our setting, where 86 percent of households are food insecure and coverage ranges from 69 to 80 percent. Cross-study comparisons therefore indicate broad magnitudes rather than precise differences in performance. Within-study comparisons of data-driven and community-based rules are more informative because both rules share the same welfare benchmark and error definitions.

The table includes every study we identified that reports a quantitative inclusion or exclusion error for a household-level targeting rule assessed against an observed welfare aggregate. We drew candidates from two reviews: a project review of community-based targeting covering 22 comparator studies, and a systematic review of targeting in fragile settings that searched Web of Science, Scopus, and ScienceDirect and extracted evidence from 135 full texts into a registry of 252 estimates. We supplemented both with reference searches from three reviews of targeting performance (Coady et al., 2004; Devereux et al., 2017; Grosh et al., eds, 2022). Six studies report errors for both data-driven and community-based rules and form the core of the table. We also include Brown et al. (2018) as a full-population benchmark for data-driven targeting. We exclude studies that assess errors against administrative eligibility or a predicted index rather than observed welfare, report only the share of all households misclassified, or measure performance solely through rank correlations, concentration shares, or targeting differentials.

The table preserves each source’s definitions and design. Schnitzer and Stoeffler (2024) match each program’s PMT selection rate to its realized community selection rate and define need as the corresponding bottom share of the welfare distribution. Their inclusion and exclusion errors therefore coincide. The reported figures are medians across nine datasets from their Table 2. They cover program areas only and omit exclusion outside those areas. Stoeffler et al. (2016) set both the consumption-based share in need and the PMT selection rate to 67 percent, matching the community caseload in their Table 5. Hillebrecht et al. (2020) define exclusion as the share of consumption-poor households not targeted, consistent with our definition. Their inclusion measure, however, is the share of consumption-nonpoor households targeted, a false-positive rate rather than one minus precision.

Basurto et al. (2020) measure the share of recipients who should not have received the subsidy. This equals exclusion error when the recipient count matches the number deemed eligible. Their PMT is simulated rather than implemented, and the figures refer to the 2012 input subsidy. Premand and Schnitzer (2021) report inclusion errors only, using consumption per capita as the welfare benchmark. Brown et al. (2018) report weighted averages across nine African countries for a basic PMT estimated using national household surveys, with selection rates of 40 and 20 percent and need defined by a fixed poverty line.

Coverage is central to interpreting these comparisons. Larger caseloads can reduce exclusion by reaching more households in need, although inclusion errors need not fall. Our coverage of 69 to 80 percent is among the highest in the table. The closest comparators are Basurto et al. (2020) at 80 percent and Stoeffler et al. (2016) at 67 percent, and our errors fall within their range. The low error levels in our setting should therefore be read alongside its high prevalence of need and broad coverage, rather than as evidence of sharper discrimination between households.

Within our study, the community-minus-PMT+ differences at realized coverage are 0.019 for inclusion error and 0.101 for exclusion error, with standard errors of 0.020 and 0.036, respectively. At equal coverage, these differences fall to 0.007 and 0.005, with standard errors of 0.007 and 0.006. These comparisons indicate that coverage accounts for much of the difference between the two rules. Among the comparison studies, only Hillebrecht et al. (2020) report standard errors.

Table C1: Targeting errors in comparative perspective: data-driven and community-based methods

Study	Country	Cov.	Bench.	Data-driven		Community	
				Incl.	Excl.	Incl.	Excl.
<i>This paper (PMT+ vs. CBT)</i>							
Realized coverage	DR Congo	80/69%	FCS	0.110	0.189	0.129	0.290
Equal coverage	DR Congo	69%	FCS	0.122	0.285	0.129	0.290
<i>Studies reporting both methods</i>							
Schnitzer and Stoeffler (2024)	Sahel, 9 prog.	45%	Cons.	0.388	0.388	0.499	0.499
<i>same programs</i>	Sahel, 9 prog.	45%	Food	0.557	0.557	0.532	0.532
Stoeffler et al. (2016)	Cameroon	67%	Cons.	0.210	0.167	0.259	0.470
Hillebrecht et al. (2020)	Burkina Faso	20%	Cons.	0.149	0.531	0.188	0.671
				(0.016)	(0.042)	(0.017)	(0.039)
Basurto et al. (2020)	Malawi	80%	Cons.	0.109	0.109	0.144	0.144
Premand and Schnitzer (2021)	Niger	40%	Cons.	0.390	–	0.420	–
<i>Data-driven only (reference)</i>							
Brown et al. (2018)	9 countries	40%	Cons.	0.310	0.360	–	–
<i>same countries</i>	9 countries	20%	Cons.	0.370	0.720	–	–

Notes: Inclusion and exclusion error rates for a data-driven rule and a community-based rule. “Cov.” is the share of households selected and “Bench.” the welfare benchmark, consumption (Cons.) or food insecurity (Food). Error definitions differ across studies and the levels are not directly comparable; the text above gives each study’s definition, source table, and denominator. Standard errors are in parentheses where the source reports them. Dashes mark measures the study does not report.

D Development of the PMT+ in Practice

This appendix documents how WFP’s Country Office (CO) developed and implemented the PMT+ rule used to assign eligibility in the status quo arm. We describe both the prediction model and the operational rules that determined the beneficiary lists. Targeting performance depends not only on model fit but also on the variables collected in the census, the eligibility threshold, and the additional criteria used to include households. The CO developed PMT+ before the evaluation began, so we evaluate an existing operational procedure.²²

²²The model-development and beneficiary-list code is maintained in WFP’s targeting workstream, separately from the evaluation pipeline. See `REPLICATION_README.md` for the file map.

D.1 Estimating the prediction model

The CO estimated the PMT using earlier WFP Post-Distribution Monitoring (PDM) surveys from the same province. These surveys measured the Food Consumption Score (FCS) and a set of candidate predictors. The CO then applied the estimated model to households in the targeting census. **[TODO: Report the PDM rounds and years, geographic restriction, and training sample size.]**

Model development proceeded in two stages. First, the CO screened candidate predictors covering household demographics, socioeconomic status, housing conditions, and asset ownership. It divided the survey data into training and testing samples containing 75 and 25 percent of observations, respectively, and compared the out-of-sample performance of cross-validated LASSO, adaptive LASSO, and forward stepwise selection. This approach draws on evidence that machine learning methods can improve targeting performance (e.g., Cao et al., 2025; Altındağ et al., 2021). **[TODO: Report the out-of-sample R^2 and MSE used to compare procedures and identify the selected procedure.]**

Second, the CO estimated an ordinary least squares regression of FCS on the retained predictors, without a constant and with heteroskedasticity-robust standard errors. The fitted values provided the household scores. The specification used to generate the beneficiary lists included the household head’s marital status, sex, age, and education category, an indicator for a single-parent household, household size, the primary income source, the number of children under five, and an indicator for any member with a disability.

Two features matter for interpretation. First, the selection procedures screened predictors, but the coefficients applied to the census came from an unpenalized regression. The final model therefore did not retain the coefficient shrinkage imposed by the penalized procedures, and its predictive performance must be assessed separately. Second, the final predictor set depended on what the census collected. Earlier specifications included asset, housing, and meal-frequency variables that were absent from the census questionnaire. The CO dropped these variables so that the model could be applied to the enumerated households. The final specification thus reflected both predictive performance and data availability. **[TODO: Consider reporting the final coefficients in an appendix table. This requires re-running the CO’s model on the training extract.]**

D.2 From predicted score to eligibility

The CO applied the model to the targeting census, collected between December 2022 and March 2023, and classified households into poor, borderline, and acceptable food consump-

tion groups using FCS thresholds of 28 and 42.²³ Households classified as poor were eligible on the basis of the model alone. **[TODO: Confirm the treatment of scores exactly equal to 28 and 42 in the implementation code and report the corresponding inequalities.]**

Only two households had missing values for a model predictor. Both were recorded in the acceptable category and subsequently included through the lock-in criteria described below. Neither was therefore excluded from assistance because of missing model inputs. **[TODO: Verify how missing predictors were handled. Distinguish a valid prediction of acceptable consumption from a missing score assigned to that category by the classification code.]**

D.3 The “Plus” component

The “Plus” component adds eligibility criteria identified through six gender-separated focus group discussions in three randomly selected blocs, as described in Section 2. Participants identified household characteristics associated with high vulnerability to food insecurity. Eight criteria were common to all three blocs. The CO adopted these shared criteria to avoid giving any single community disproportionate influence over the rule.

The evaluation report lists the following eight criteria:

1. Households headed by a woman with no other member engaged in an income-generating activity.
2. Households headed by a person aged 60 or older.
3. Households headed by a person with a disability.
4. Households headed by a person with a chronic illness.
5. Households with orphaned children.
6. Households with a pregnant or lactating woman.
7. Households headed by a widower.
8. Displaced households.

A household meeting at least one lock-in criterion was eligible regardless of its predicted FCS. These criteria could therefore only expand eligibility. They increased the number of selected households by 29 percent relative to the model alone.

²³Section 3.1 discusses the FCS and the use of 28 as the food-insecurity threshold.

The census also constrained which community criteria could be used. Each criterion had to correspond to a collected variable. For example, the presence of a malnourished child was commonly proposed but could not be implemented because the census did not measure it. **[TODO: Reconcile the three lists of lock-in criteria. The list above follows WFP’s evaluation report. Section 2 combines disability and chronic illness and adds adolescent motherhood. The implementation code applies seven criteria, omitting widowerhood. Confirm which criteria generated the beneficiary lists and align the descriptions.]**

D.4 Operational adjustments

Two additional rules affected the scope of household targeting.

WFP’s standard operating procedures call for blanket coverage when more than 90 per cent of a population is eligible, because excluding the remaining minority may not justify the administrative cost. PMT+ coverage in the study blocs remained below this threshold, so household targeting proceeded as planned.

Separately, in April 2023 the CO decided to provide blanket coverage in three internally displaced person camps. These blocs are excluded from the analysis sample. **[TODO: Verify whether the three blanket-covered IDP blocs were outside the 96 eligible study blocs and clarify their treatment in the sample description.]**

E Ethics, Consent, and Pre-registration

The study was conducted in partnership with the World Food Programme as an evaluation of an operational assistance program. Enumerators obtained informed consent from each respondent before every survey round, explaining the purpose of the research, the voluntary nature of participation, and the confidentiality of responses; participation in the survey was independent of eligibility for assistance. Data were stored and analyzed in de-identified form. The impact-evaluation design received ethical approval from the Institutional Review Board of the University of Milan-Bicocca (approval no. 759), and the study adhered to the 2020 United Nations Evaluation Group (UNEG) ethical standards under the oversight of WFP’s Office of Evaluation and the World Bank’s Development Impact (DIME) department.

The evaluation was pre-registered with the American Economic Association RCT Registry on July 20, 2023 (AEARCTR-0010350).²⁴ The registry names targeting precision—

²⁴Registered as “Testing Targeting Variants for Emergency Transfers in DRC,” <https://www.socialscisceregistry.org/trials/10350>.

inclusion and exclusion errors—and satisfaction with the targeting process as primary outcomes; the equal-coverage simulation and the mechanism analyses are exploratory extensions developed in the course of the study.